

Large Language Model Powered Knowledge Graph Construction for Mental Health Exploration

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors propose to use GPT model to automatically construct a KG about mental health. The authors demonstrate that this KG is very useful for mental health related tasks on a UKBB MDD prediction problem. It is an interesting approach and the resulting KG could be very useful. I also like the active learning approach to identify useful abstracts. The conditional statement is also interesting and important.

A few questions:

1. Can this procedure to be generalized to other diseases? Is it possible to generate a context-specific KG automatically?
2. How to control the noise in the KG? Especially for the literature based method, there should be lots of noise. Also, what if there is contradictions among two statements?
3. Line 263, advanced entity alignment and linking techniques -> what are they?
4. The results section need to be more precise and specific. What exactly is the experiment that shows improved performance? Currently, the result is given but the experimental setup is highly obscure to judge if the result is reasonable or solid.
5. Have you compared with GraphRAG? It also takes in a set of documents and create a KG.
6. The texts in Figure2a are very difficult to see.
7. Have you compared with the knowledge graph constructed using alternative LLMs? For example, GPT3 or claude? Would the result look worse? That would be an interesting.
8. Could you provide cost analysis of GPT4 model in constructing the KG?
9. Have you also compared with traditional noun-entity recognition approach to construct KG in contrast to GPT4? It would be interesting to see performance difference.
10. For the MDD study, how is the 2000 individual control chosen? Also, why choose MDD? This procedure seems to be standardized across other mental health disorders as well - to show robustness, maybe the authors could experiment on other diseases as well.
11. I feel like the approach to represent patient could be explained more clearly. Why SapBERT? Why a simple aggregation of frozen KG embeddings? Can it learn end-to-end using graph neural networks?

12. Why is air pollution an important factor for MDD?

13. Why is non-medical EHR embedding have such a good performance? Is there potential confounders? How do you select these variables to avoid confounders?

14. How do the authors plan to visualize the KG and deliver to the users? Currently, the github is not available.

(Remarks on code availability)

The code is not available at the github site.

Reviewer #2

(Remarks to the Author)

I would like to start by commending the authors for their substantial effort in tackling a highly complex and rapidly evolving field. The integration of a variety of advanced techniques—including text extraction, data formatting, web and database crawling, and visualization—is no small feat. Structuring this breadth of work into a coherent and readable format is undeniably challenging. However, I believe that only an extensive revision, if at all possible, could make the findings of this study truly beneficial to the reader. My primary concerns center around the use of non-open-source systems, such as ChatGPT, and the absence of rigorous validation processes for the conclusions drawn from the knowledge graph. While the authors mention cross-checking 300 articles, this seems to be a minimal fraction of the data processed, leaving the reliability of the results in question.

1. Use of ChatGPT: While ChatGPT demonstrates impressive performance across many benchmarks, the choice of a closed, proprietary platform for constructing a shared knowledge graph is problematic. A key expectation for a knowledge graph project is that it should be built on an openly available large language model (LLM), rather than a closed system like OpenAI's platform. The use of closed systems hinders transparency in understanding the deeper structures of the knowledge graph, raising concerns about the potential for data misuse by third-party providers.

2. Reliability of Insights: There is a lack of clarity regarding how users can verify the reliability of insights derived from the knowledge graph, particularly those not covered by established medical databases. It is crucial when employing LLMs that users are encouraged to critically assess machine-generated content rather than blindly trust it. Hallucinations—a known issue in many domains involving LLMs—pose a specific risk. How has this been addressed or mitigated in the current implementation?

3. Lack of Comparable Improvements: While the authors claim that the inclusion of "side facts" in their knowledge graph is a key advantage, the report does not provide sufficient details for comparing this system with existing knowledge graphs. Moreover, the heavy reliance on literature (over 90%) in constructing the graph casts doubt on the robustness of their approach, particularly with respect to the integration of databases.

4. Superficial Examples: The examples provided in the study remain superficial, with many of the associations being either obvious or, in some cases, misinterpreted. For instance, an association between two entities does not automatically imply risk or causality. The examples presented, such as the one involving anxiety, could easily be found by any researcher conducting a PubMed search, which raises the question of what unique benefit this knowledge graph offers.

5. Need for Thorough Analysis: The concept of an enriched knowledge graph is indeed promising, but a more thorough analysis of the various stages involved is needed. The rapid pace of developments in this field does not excuse the lack of rigorous validation at each step of the process. This is particularly relevant when it comes to the UK Biobank analysis, where the methodology for using the knowledge graph in predictions remains entirely unclear. In fact, the scope of the current work seems too broad, and I would recommend splitting the project into two separate works: one that focuses on the in-depth analysis and validation of the knowledge graph itself, and another dedicated to the UK Biobank project.

(Remarks on code availability)

Although we tried several times, it was not possible to access the github code!

Reviewer #3

(Remarks to the Author)

In this manuscript the authors have developed a mental disorders knowledge graph. The authors have used both large language models (LLMs) to extract medical relations from PubMed abstracts, as well as existing biomedical knowledge bases for the relations. They have showcased the potential of this knowledge graph for predicting mental health disorders in UK BioBank data. The methodology to develop the knowledge graph is sound, however, the organization of content in the manuscript can be improved.

Major questions that will require more clarity:

- Why have the authors limited the extraction to just PubMed abstracts and not included the entire articles from the PubMed Central corpora? There seems to also be a discrepancy in the number of abstracts used, since the discussion mentions 500, but the online methods section mentions 234,000+ articles.

- Why have the authors focused on the use of PheCode over ICD-10? The authors include a line in the manuscript, however it needs more explanation. For an application of this research, would not prediction on ICD-10 codes be more meaningful?

- The authors have mentioned that the MDKG is more extensive than its counterpart knowledge graphs (e.g., PrimeKG or SPOKE), however, having more relations in a graph does not necessarily mean higher quality or usefulness. It would have been useful to:

--- Either see the added benefit of using MDKG for prediction tasks over BioBank, when compared with knowledge embeddings from SPOKE or PrimeKG or GENA (i.e., SVM + GENA AUROC).

--- OR do a human evaluation of random relations from MDKG vs GENA vs PrimeKG

- Why have the authors used QuickUMLS over CTAKES or MetaMap?

- Section 2.5.2 Scenario 3 must indicate the AUROC over all variables + KG embeddings. The AUROC for SVM (all variables) + KG and LR (all variables) + KG are mentioned in the figure, but not in the results section. I am curious why the authors have focused on SVM and LR models instead of models such as XGBoost or Random forests?

- Regarding SHAP analysis: Is there a way to indicate what is captured in certain embedding dimensions to be determined of higher predictive power?

- Have the authors used ChatGPT interface or the underlying API (e.g., GPT 3.5)? Were there any differences between ChatGPT and GPT-4o? What are the added benefits for entity annotation using QuickUMLS that is not captured through the use of LLMs?

- Capturing contextual information in knowledge graphs is not entirely novel has been a topic that has been covered in other publications. What is the underlying model used by the authors (e.g., RDF*, reification)

(Remarks on code availability)

The URL for the Github repository does not work and it shows 404 not found error.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed my concerns adequately. I don't have further comments.

(Remarks on code availability)

Tutorials and demos would be beneficial.

A UI interface like gradio would be useful and increase the impact.

A python API would also be nice.

Reviewer #2

(Remarks to the Author)

The revised manuscript significantly improved the clarity and overall quality, addressing several issues effectively. However, a few minor points remain unclear or insufficiently justified:

1. Usage of Phecodes vs. ICD-10:

The explanation provided for choosing phecodes over ICD-10 codes remains unconvincing. The authors argue phecodes avoid confusion with billing practices inherent to ICD-10. However, a simpler approach to reducing entities and complexity could be using the first two digits of ICD-10 codes, thereby achieving the intended simplification without introducing an entirely different coding system. The authors should provide a clearer justification for preferring phecodes specifically.

2. Choice of ChatGPT (Closed-source) Over Open-source Alternatives:

The authors still have not sufficiently clarified their rationale for selecting ChatGPT, a proprietary closed-source model, over available open-source alternatives such as BERT or Llama. Given the sensitivity of medical data and the importance of transparency, employing a locally hosted or open-source system seems preferable. The authors should clearly justify the benefit or necessity of their current choice.

3. Clarification of "Disease Prediction":

The use of the term "disease prediction" remains ambiguous. It is unclear if the authors are predicting future diagnoses or simply identifying patients already labeled with Major Depressive Disorder (MDD) in their dataset. The authors must explicitly clarify if their approach involves genuine predictive modeling—predicting future onset of disease—or merely labeling existing diagnoses accurately.

4. Explanation and Usage of "Triplets":

While "triplets" are now mentioned in the abstract, the manuscript lacks a specific and clear definition of what constitutes a

"triplet" and the rationale behind their use. Providing a concise explanation in the methodology would enhance clarity and understanding of their importance within the knowledge graph.

5. Clarification Regarding AUC Values:

The authors mention an improvement in AUC values (abstract) "from 0.02 to 0.21." This statement remains unclear and potentially misleading. Typically, an AUC (Area Under the Curve) value below 0.5 indicates a model performing worse than random chance, with meaningful AUC values typically ranging from 0.5 (random chance) to 1.0 (perfect classification). Thus, an AUC improvement to 0.21 is confusing and suggests either a typographical error or a conceptual misunderstanding. The authors should review and clarify this to avoid misinterpretation.

Addressing these points will further improve the manuscript's precision, clarity, and overall impact.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

Thank you to the authors for the detailed rebuttal, and updated analyses and results.

The authors have addressed all my feedback and concerns from the previous round. I have one final concern, that is outlined below:

- While the SVM / LR results are acceptable, and the authors have also included results from XGBoost and Random Forest models in the recent iteration, I am confused on the claim around interpretability - primarily around the response for model selection and the SHAP analysis.

That is, the following statement in the rebuttal: "We selected SVM and LR models as our primary approaches due to their interpretability and simplicity" is contradictory to another statement: "Given the inherent instability of embeddings—meaning that feature correspondence across different training runs is not fixed—interpreting individual dimensions remains a challenge."

Why have the authors not pursued the use of deep learning models for this task when interpretability and justifications on the inference, are going to be impossible even with SVM / LR models?

(Remarks on code availability)

The authors have provided this Github repository <https://github.com/MonsterTea/MDKG?tab=readme-ov-file> that has a detailed README section on execution of Python code. There are some minor typos in the README that could prohibit other users from running the code correctly.

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Response for Reviewer 1

1. *Can this procedure to be generalized to other diseases? Is it possible to generate a context-specific KG automatically?*

Response: Thank you for your insightful question. Yes, this procedure can be generalized to other diseases, and a context-specific knowledge graph (KG) can be generated automatically. However, several challenges must be considered. Each disease may have distinct pathological pathways, biomarkers, and other critical features. Without carefully designing disease-specific entities, relationships, and contextual attributes, an automatically generated KG may fail to fully capture all relevant information. While current automated methods are more effective for constructing general-purpose KGs, advancements in large language models (LLMs) offer promising potential. By processing and analyzing vast amounts of literature, LLMs can emulate the reasoning involved in human KG design, paving the way for more targeted and disease-specific KGs in the future.

2. *How to control the noise in the KG? Especially for the literature-based method, there should be lots of noise. Also, what if there is contradictions among two statements?*

Response for "How to control the noise in the KG":

Thank you for raising this important question. Controlling noise in a knowledge graph (KG), particularly when sourcing information from literature, is a significant challenge. We have dedicated considerable effort to mitigating noise in our work and ensuring that users have insights into confidence levels when interpreting the data.

- **Technical Noise Control:** To reduce noise, we employ a multi-step refinement process for the extracted triplets. First, entities and relationships are identified using a BERT-based model. Then, GPT-4o is used to refine these triplets through a structured validation process that includes: (i) Verifying whether entity spans are correctly identified. (ii) Ensuring that the entities are of the correct type. (iii) Validating the relationship between the identified entities. and (iv) Assigning a confidence score (1-10) to assess the overall accuracy of the triplet. Triplets scoring below 5 undergo further refinement to enhance their accuracy. This iterative approach helps ensure that the extracted knowledge is more reliable and meaningful.
- **Noise in the Literature:** The inherent noise in literature is complex and challenging to eliminate, as even well-established findings can evolve over time. Many studies may not always yield universally valid or reliable conclusions due to variations in sample size, demographic factors, experimental design, statistical methods, and study origins. To address this, we incorporate several contextual features when

extracting triplets from the literature, including sample size, population demographics (e.g., race, gender, age), data source details (e.g., publication date, journal, impact factor), publication type, and statistical methodologies used. By providing this additional context, we enable users to better assess the reliability and relevance of the extracted knowledge.

- **Metrics to Reflect the Reliability and Confidence:** In addition to these measures, we designed two additional metrics to reflect the reliability and confidence of the triplets:

Frequency: For each triplet, we track its frequency across all sources (both literature and integrated existing databases). A higher frequency indicates that the triplet appears in multiple sources, suggesting greater reliability.

Confidence Score: We calculate a confidence score for each triplet derived from literature, which is based on several factors including model prediction scores, journal impact, frequency across sources, distinct publication years count, max publication year, and contradiction counts in the data. By integrating these factors into the confidence score, we provide a quantitative measure of reliability, helping users assess the quality of the insights derived from the knowledge graph. For triplets with high confidence scores, users can be more confident in their accuracy, while triplets with lower scores may require further evaluation. This approach enables us to reduce the noise in the knowledge graph and provide users with tools to assess the reliability of the information. Additionally, through manual evaluation, we found a strong positive correlation between our confidence score and human-assigned evaluation scores (mean correlation: 0.78), further validating the effectiveness of our confidence score.

Response for "contradictions among two statements":

Indeed, contradictions between statements can arise. In our KG, such contradictions are restricted to risk and therapeutic relationships. For instance, if entity A is a risk factor for entity B , it should not simultaneously be the treatment for entity B . To address these contradictions, we employ two approaches:

1. For triplets exhibiting contradictions, we provide both the conflicting triplets and the corresponding full abstracts to GPT-4o. The model determines whether the contradictions stem from differing conclusions across the articles. If so, we retain the conflicting triplets to reflect the variability in the literature. If not, the model selects the correct triplet, and we remove the incorrect one from our KG.
2. We also introduce a penalty for contradictions in the confidence score to account for these inconsistencies.

These measures help ensure that our KG maintains consistency while preserving the complexity of real-world scientific findings.

[Change Made in Online Methods 4.2.1 Named Entities, Relationships, and Conditional Statements Extraction]

After extracting entities and relationships with the BERT-based model, we refined the extracted triplets using GPT-4o. The refinement process includes verifying entity

spans, confirming entity types, validating relationships, and scoring each triplet (1-10) for overall accuracy. Triplets with a score below 5 are reprocessed. In cases of contradictory statements, we input the conflicting triplets along with the full abstract into GPT-4o to determine whether the contradictions stem from differing conclusions in the articles. If the contradictions are due to differing conclusions, they are retained; otherwise, the model selects the correct triplet, removing the incorrect one from the knowledge graph.

The best-performing model demonstrates high accuracy in entity recognition, achieving a micro F1 score of 0.89 and a macro F1 score of 0.88, indicating reliable precision across different entity types. Relationship extraction presents more challenges, with initial micro and macro F1 scores of 0.68 and 0.67, respectively. When both tasks are performed jointly, the model achieves micro and macro F1 scores of 0.63 and 0.63, highlighting the complexity of integrating these processes. However, following the post-processing refinement steps, significant improvements are observed, with the joint model’s micro F1 score rising to 0.79 and the macro F1 score increasing to 0.76, demonstrating a marked enhancement in overall performance.

[Change Made in Main Text 2.1.2 Incorporation of Contextual Information]

To assess the credibility of triplets within MDKG, we incorporate multiple factors that influence their confidence levels. Reliability is first evaluated through cross-source consensus, where triplets appearing in multiple sources are considered more trustworthy. For literature-derived triplets, we further document key attributes such as publication type (e.g., reviews, clinical trials, meta-analyses) and analytical methods used (e.g., multivariate analysis, multiple regression, logistic regression), as these factors contribute to varying levels of evidentiary strength. To enhance reliability assessment, we assign a confidence score (detailed in Appendix C) to each triplet. This score integrates multiple factors, including model prediction scores, journal impact, frequency across sources, distinct publication years count, maximum publication year, and contradiction counts in the data. By appending this confidence score to each triplet, MDKG enables users to prioritize high-quality data and better assess the reliability of the extracted knowledge.

[Change Made in Appendix C]

Confidence Score Calculation

The confidence score integrates multiple quality indicators of scientific statements (triples) using a weighted framework. Each component score is normalized to [0,1] and scaled by a predefined weight, ensuring the final score remains within the range [0,3]. Prior research on assessing study quality has highlighted key factors influencing a study’s credibility and impact, including impact factor, statement frequency, and the number of distinct publication years^[1-3]. Our framework incorporates these established factors and further refines their contributions based on insights from domain experts, ensuring a balanced and context-aware evaluation. The confidence score is computed

118 as follows:

$$S_{confidence} = S_{prediction} + 0.8 \cdot S_{IF} + 0.6 \cdot S_{Frequency} + 0.4 \cdot S_{years} + 0.2 \cdot S_{date} - 0.5 \cdot S_{contradiction}$$

- 119 • **Model Prediction Score** ($S_{prediction}$): Quantifies the base trustworthiness of the
 120 triple generated by the prediction model. The model directly outputs the score,
 121 ranging from $[0, 1]$.
- **Impact Factor Score** (S_{IF}): Reflects the academic influence of the resource.

$$S_{IF}(i) = 1 - \exp(-\alpha i)$$

122 where i is the average impact factor of the source journal. To align with the 75th
 123 percentile impact factor value (6.6), the decay coefficient $\alpha = 0.25$ ensures $S_{IF} \geq 0.8$
 124 when $i \geq 6.6$.

- **Frequency Score** ($S_{Frequency}$): Measures how frequently the triple appears across different resources.

$$S_{IF}(n) = 1 - \exp(-\alpha(n-1))$$

125 where n is the occurrence count of the triple. With the 75th percentile $n = 4$, $\alpha =$
 126 0.55 ensures $S_{Frequency} \geq 0.8$ when $n \geq 4$.

- **Distinct Publication Years Counts Score** (S_{years}): Captures the historical consistency of the triple across years.

$$S_{years}(n_t) = 1 - \exp(-\alpha n_t)$$

127 where n_t is the number of distinct publication years. For the 75th percentile $n_t =$
 128 4 , $\alpha = 0.4$ yields $S_{years} \geq 0.8$ when $n_t \geq 4$.

- **Max Publication Year Score** (S_{date}): Rewards recency of the triple's latest publication.

$$\begin{cases} S_{date}(t) = \exp(-\alpha(T-t)), & t \leq T \\ S_{date}(t) = 1, & t > T \end{cases}$$

129 where T is the reference year (e.g., 2025) and t is the latest publication year. The
 130 Decay Rate $\alpha = 0.1$ imposes gradual penalties for older publications.

- **Contradiction Counts Score** ($S_{contradiction}$): Penalizes conflicting statements from other sources.

$$S_{contradiction}(c) = \exp(-\alpha c)$$

131 where c is the count of contradictory triples. $\alpha = 0.1$ ensures a penalty of 0.8 when
 132 $c = 2$.

133 In summary, the confidence score aggregates multiple aspects of a resource's qual-
 134 ity and relevance, with each factor having a distinct role in shaping the final score.
 135 The decay functions ensure that the score is adjusted for diminishing returns, while
 136 the weights reflect the relative importance of each factor in the overall assessment.

137 This approach allows for a nuanced evaluation of resources, balancing credibility, aca-
 138 demic impact, historical significance, recency, and frequency of use. The distribution
 139 of confidence scores across different types of relationships is illustrated in Figure 1.

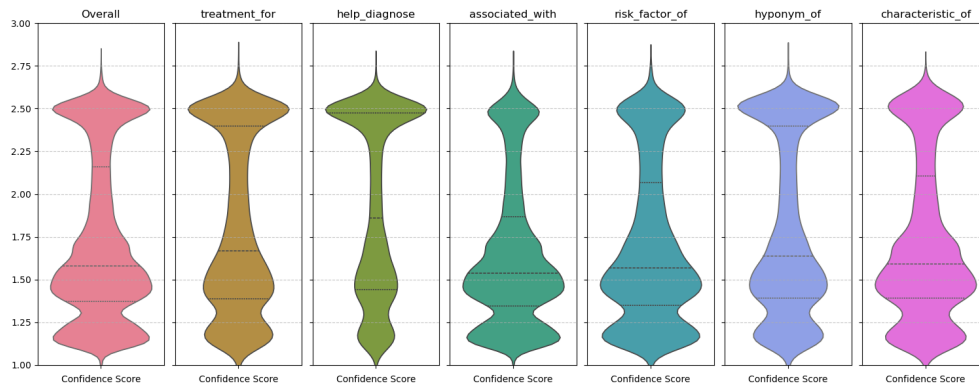


Fig. 1 The distribution of confidence scores for different relationships.

140 3. Line 263, advanced entity alignment and linking techniques – > what are they?

141 **Response:** Thank you for your comment. The "advanced entity alignment and link-
 142 ing techniques" mentioned in the Results section are indeed detailed in the Methods
 143 Section 4.3 under Entity and Relationship Alignment. In that section, we describe
 144 our approach to aligning entities and relationships across various biomedical sources,
 145 which includes:

- 146 • **Disambiguation of Abbreviations:** We leverage the "abbreviation for" relation-
 147 ship to resolve abbreviations, ensuring their correct interpretation and linkage.
- 148 • **Linking to Biomedical Databases:** Using SapBert embeddings and the FAISS
 149 library, we identify the closest entity matches and link them to established
 150 biomedical databases such as UMLS, MONDO, UBERON, and Gene Ontology.
- 151 • **Clustering Unlinked Entities:** For entities that cannot be directly linked to spe-
 152 cific biomedical databases, we use clustering based on SapBert embeddings. The
 153 most frequently occurring entity within each cluster is designated as the representa-
 154 tive or "standard" entity. To ensure proper recognition and integration within our
 155 MDKG, we assign a unique ID to each cluster.

156 We hope this clarifies the methodology behind our entity alignment and linking pro-
 157 cess. To enhance clarity, we have revised the manuscript accordingly. The updated
 158 sentence now reads:

159 [Change made in Main Text 2.3 Integration and Synthesis of MDKG]

160

161 Using systematic entity alignment and linking techniques (details in online Methods
162 Section 4.3), we merge this curated data with the literature-based MDKG...

163 4. *The results section need to be more precise and specific. What exactly is the*
164 *experiment that shows improved performance? Currently, the result is given but the*
165 *experimental setup is highly obscure to judge if the result is reasonable or solid.*

166 **Response:** Thank you for your valuable suggestion. We have revised the Results
167 section to provide more detailed insights into the experiments demonstrating improved
168 performance. The key updates are as follows:

169 **1. Human Evaluation of Triplet Quality Across Knowledge Graphs:** We have
170 clarified the human evaluation results, which indicate that our MDKG achieves a
171 high knowledge accuracy of 79.0%. Our confidence score design showed a strong
172 positive correlation with evaluator assessments of triplet quality (mean correlation:
173 0.78, standard deviation: 0.00), demonstrating the reliability and effectiveness of
174 our scoring approach. Additionally, we compared the average time evaluators spent
175 assessing triplets across different knowledge graphs. MDKG’s rich contextual fea-
176 tures significantly reduced evaluation time by 62–77% compared to PrimeKG and
177 GENA, further validating the practical utility of our contextual enhancements.

178 **2. Disease Prediction with Knowledge Graph Embeddings:** We have expanded
179 the disease prediction analysis by incorporating two additional classification mod-
180 els (Random Forest and XGBoost) and two new mental health disorders (Anxiety
181 and Bipolar Disorder). Results consistently demonstrate that integrating medical
182 knowledge graph embeddings enhances predictive accuracy across all tested disor-
183 ders. These additional analyses provide stronger evidence supporting the value of
184 integrating knowledge graph embeddings into disease prediction models.

185 We hope these refinements address your concerns and further strengthen the case for
186 our knowledge graph’s effectiveness. Below, we have included the revised content from
187 Human Evaluation, and the updates from Disease Prediction can be found in response
188 to your comment on question 10.

189 **[Change Made in Main Text 2.4 Human Evaluation of Triplet Quality**
190 **Across Knowledge Graphs]**

191 We conducted a comparative analysis of MDKG against two large-scale general
192 biomedical knowledge graphs—PrimeKG [4] and SPOKE [5]—as well as a mental
193 health-specific KG, GENA [6]. Our focus was on the density of directly connected
194 information relevant to MDD, anxiety, bipolar disorder, schizophrenia, and depres-
195 sion. The analysis revealed that MDKG contains 5 to 50 times more information on
196 these mental disorders than the comparative KGs, as shown in Figure 4d, highlighting
197 its superior coverage and comprehensiveness.

198 To further assess MDKG’s quality, we conducted a human evaluation of randomly
199 selected relationships from PrimeKG, MDKG, and GENA. Four evaluators—two psy-
200 chiatrists (d1, d2) and two medical PhDs (phd1, phd2)—assessed the correctness and
201 relevance of the extracted triplets. Given the varying information richness of the three
202 knowledge graphs, we provided the maximum available data for each: triplets, PMIDs,

and sentences for GENA; triplets and database sources for PrimeKG; and triplets, PMIDs, sentences, journal, impact factor, frequency, and date for MDKG. For evaluation, we selected two diseases, schizophrenia and anxiety disorders, and randomly sampled 100 non-redundant triplets per disease. Each evaluator reviewed the same set of examples, ensuring a consistent and unbiased comparison.

Evaluation metrics included correctness (whether the triplet was correct based on the provided sentence), insightfulness (rated 1–3 for novelty, reliability, and usefulness), and evaluation time (average time for 200 examples). Correctness was excluded for PrimeKG, as it lacks original sources. Data analysis used Cohen’s Kappa for inter-rater consistency, with values ranging from -1 (no agreement) to 1 (perfect agreement), and the agreement rate between evaluators. Following is a summary of our findings:

- **Comparative Analysis of Triplet Distributions:** Overall, for triplets related to anxiety and schizophrenia, the relation and entity architecture across the knowledge graphs demonstrate distinct graph differences, as shown in Figure 2. PrimeKG predominantly contains generic associations, with approximately 90% of the relations labeled as "associated with," mostly involving gene entities. In contrast, MDKG exhibits more clinically actionable relationships, including approximately 20% "therapeutic relationships," 11% "risk relationships," and 10% "diagnostic relationships." GENA’s relationship distribution demonstrates linguistic specificity but clinical fragmentation, with approximately 85% of relations occurring 5 times. As a result, many of these relations lack standardization, such as "RECEIVE APPROVAL BY U.S. FOOD AND DRUG ADMINISTRATION FOR TREATMENT OF."
- **Quality Metric Performance:** For correctness (Figure 3b), MDKG achieved an accuracy of 79.0% (SEM = $\pm 0.07\%$), slightly outperforming GENA, which reached 78.3% (SEM = $\pm 0.54\%$). The errors in MDKG were primarily attributed to semantic ambiguities (28%), rather than factual inaccuracies (12%). In terms of insightfulness, PrimeKG (2.09 ± 0.01) and MDKG (2.18 ± 0.02) scored lower than GENA (2.29 ± 0.02). This difference likely reflects evaluators’ emphasis on the novelty of knowledge as a key metric. While PrimeKG and MDKG offer more extensive knowledge integration—especially MDKG, which incorporates clinical consensus data dating back to 1985—24.34% of MDKG’s schizophrenia-related triplets were derived from studies published before 2010. Evaluators frequently described these entries as "consolidated textbook knowledge" in their post-evaluation feedback. This observation highlights the challenge of the need to balance evidence maturity (strength) with temporal relevance (novelty) in KG development.
- **Inter-Rater Reliability Assessment:** We observed significant intra-group vs inter-group discrepancies in evaluation consistency (Figure 3e). Clinical experts (d1-d2) achieved moderate agreement in correctness assessment (kappa: 0.43, Agreement: 76.7%), substantially higher than cross-disciplinary pairs like d2-phd1 (kappa: 0.04). Similarly, researchers (phd1-phd2) showed strong consensus in insightfulness evaluation (kappa: 0.63, Agreement: 80.0%), whereas clinician-researcher pairs had a kappa value of only 0.28. This pattern suggests divergent evaluation priorities—clinicians may have prioritized therapeutic validity (e.g., contraindications, treatment effects), while researchers placed greater emphasis on mechanistic explanations.

- **Time Cost of Evaluation:** MDKG’s enriched contextual features significantly reduced evaluation time by 62–77% compared to PrimeKG and GENA (Figure 3c). Clinicians spent an average of 31 ± 2.24 seconds per MDKG triplet, compared to 129 ± 11.51 seconds for PrimeKG and 85.25 ± 4.66 seconds for GENA. This reduction in evaluation time was strongly correlated with metadata completeness, highlighting MDKG’s efficiency in facilitating rapid and informed assessments.
- **Validation of Confidence Metric:** Our confidence score exhibited a strong positive correlation with human insightfulness ratings (mean correlation: 0.78, std: 0.00), indicating that the factors used in its design closely align with the criteria humans consider when assessing the value of a triplet. This strong alignment further validates the robustness and reliability of our approach.

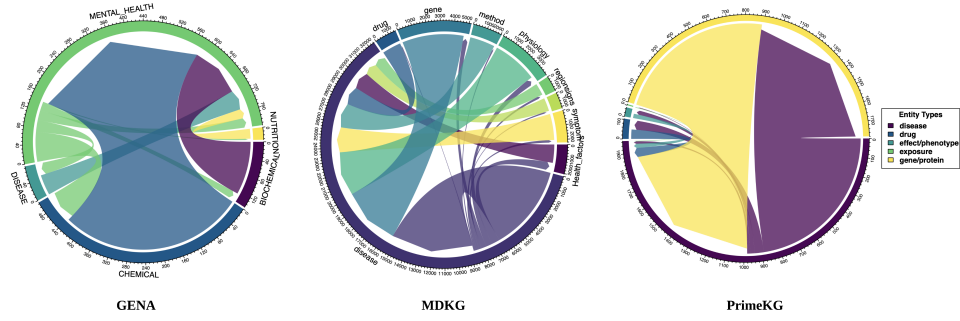


Fig. 2 Connections between different entity categories across different knowledge graphs.

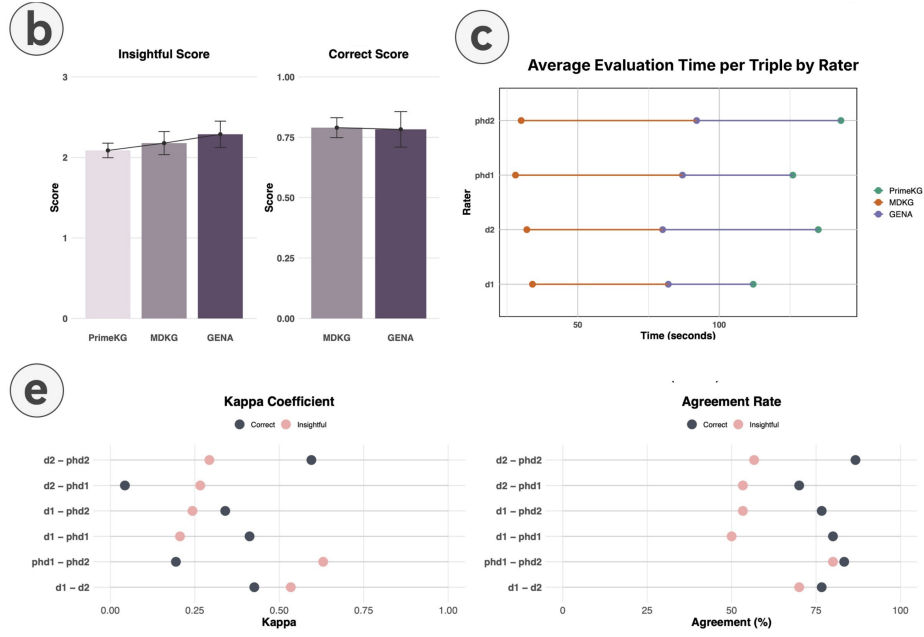


Fig. 3 The Manual check Results.

259 5. Have you compared with GraphRAG? It also takes in a set of documents and create
260 a KG.

261 **Response:** Thank you for your insightful question! There are several key differences
262 between our approach and GraphRAG, as summarized in Table 1.

263 In essence, GraphRAG represents an advanced paradigm of Retrieval-Augmented
264 Generation (RAG) technology, designed to optimize retrieval efficiency for specific
265 tasks by dynamically constructing subgraphs from a knowledge graph[7]. Its primary
266 goal is to enhance the performance of large models in applications such as question
267 answering and summarization.

268 In contrast, traditional knowledge graphs focus on building persistent, compre-
269 hensive knowledge networks that serve as foundational resources for a wide range
270 of applications. This structural difference makes traditional knowledge graphs more
271 suitable for long-term, large-scale projects, whereas GraphRAG is better suited for
272 smaller, task-specific applications that require dynamic information retrieval.

273 We hope this explanation clarifies the differences between our method and
274 GraphRAG while highlighting the advantages of our approach in developing a more
275 complete and versatile knowledge graph.

276 6. The texts in Figure2a are very difficult to see.

Table 1 Comparison between GraphRAG and Our Framework

Dimension	GraphRAG	Our Framework (Traditional KG)
Document Processing Paradigm	(i) Pre-Construction Filtering: Filters documents upfront based on embedding similarity. (e.g., retaining only top 10% relevant docs) (ii) Irreversible Filtering: Critical concepts may be permanently excluded if not deemed similar enough during preprocessing. (iii) Query-oriented retrieval: Optimized for 1-2 hop relationships	(i) Inclusive Initialization: Incorporates all documents into the initial construction phase. (ii) Post-Construction Refinement: After obtaining the KG, subgraphs of interest can be obtained through conditional filtering.
Architectural Focus	Query-Driven Design: Builds task-specific graphs optimized for immediate question-answering performance. <i>Therefore, graph design can be tailored to the problems</i>	(i) Domain-Centric Design: Constructs a unified and complete knowledge graph (e.g., a mental health KG with over 1.7M nodes). <i>A predefined KG framework may not fully meet all requirements.</i> (ii) Persistent Storage: Maintains all extracted relationships, nodes, and contextual features as a research-ready database.
Update Mechanism	Session-level updates (subgraphs rebuilt per query)	Versioned updates (daily/weekly iterations)
Task Adaptability	Primarily optimized for question answering and similar tasks by constructing focused subgraphs.	Multi-Task Readiness: Naturally supports operations like link prediction, knowledge search, multi-hop reasoning, and other cross-task functionalities due to its complete and flexible knowledge representation.

277 **Response:** Thank you for your valuable feedback. We have revised Figure 2a (now
278 Figure 4) by increasing the font size and removing relationship pairs with very low
279 proportions to enhance readability.



Fig. 4 Relationship Distribution in MDKG

7. Have you compared with the knowledge graph constructed using alternative LLMs? For example, GPT3 or claude? Would the result look worse? That would be an interesting.

Response: We appreciate the reviewer’s suggestion to compare the knowledge graph constructed using GPT-4 with alternative LLMs, such as GPT-3, ChatGPT, and Claude. While this comparison was not included in the original article, we have now conducted a comparison of these models using Entity Recognition and Relationship Extraction tasks.

[Change made in Appendix D.4]

Comparative Analysis of Large Language Models for Biomedical Information Extraction

We systematically evaluated the performance of four language models—GPT-4 (gpt-4), GPT-3 (text-davinci-003), Claude-3-opus, and ChatGPT (GPT-3.5-turbo-1106)—on two key biomedical natural language processing (NLP) tasks: named entity recognition (NER) and relationship extraction (RE). Both tasks were formulated as multi-class classification problems. To benchmark performance, we used MDIEC, a dataset of 500 manually annotated medical abstracts from our institutional corpus.

The dataset included nine entity types (Anatomy, Disease/Symptom, Drug/Chemical, Gene/Protein, Health Factors, Method/Procedure, Phenotype, Physiology, and Sign) and seven relationship types (Therapeutic, Diagnostic, Risk, Characteristic, Association, Hierarchical, and Abbreviation). A standardized few-shot learning framework was applied, and model performance was evaluated using precision, recall, and F1-score metrics.

For NER (Table 2), Claude-3-opus demonstrated marginally superior performance to GPT-4 (F1: 0.87 vs. 0.85). In relationship extraction (Table 3), GPT-4 achieved superior performance (F1: 0.66 vs. Claude-3-opus: 0.63). Both models significantly outperformed GPT-3.5-turbo (NER F1: 0.74; RE F1: 0.55) and GPT-3 (NER F1: 0.63; RE F1: 0.51) across all metrics.

Table 2 Comparative Analysis of LLMs in Entity Recognition Task)

Model	Precision	Recall	F1-score
Claude-3-opus	0.89	0.85	0.87
GPT3	0.67	0.59	0.63
ChatGPT	0.71	0.78	0.74
GPT-4	0.87	0.84	0.85

Table 3 Comparative Analysis of LLMs in Relationship Extraction Task

Model	Precision	Recall	F1-score
Claude-3-opus	0.64	0.62	0.63
GPT3	0.45	0.40	0.42
ChatGPT	0.57	0.54	0.55
GPT-4	0.67	0.64	0.66

8. Could you provide cost analysis of GPT4 model in constructing the KG?

Response: Thank you for your request regarding the cost analysis of using GPT-4 in constructing the knowledge graph. Below is a breakdown of the associated costs:

- **Use of GPT-4 and GPT4o:** Annotation of 500 abstracts (for entity recognition and relationship extraction) with GPT-4 cost approximately \$120. The GPT-4 model used for this task was priced at \$10 per 1 million tokens (input) and \$30 per 1 million tokens (output). Validation of 2 million triplets for accuracy using GPT-4o costs about \$800, with the price being \$2.50 per 1 million tokens (input) and \$10 per 1 million tokens (output).
- **Use of ChatGPT:** For simpler tasks such as table extraction and cohort information extraction, we utilized GPT-3.5, which provided a more cost-effective solution

320 while maintaining efficiency. This cost analysis highlights our approach to opti-
321 mizing expenses by selecting models based on task complexity and computational
322 requirements. Let us know if you need further details!

323 *9. Have you also compared with traditional noun-entity recognition approach to con-*
324 *struct KG in contrast to GPT4? It would be interesting to see performance difference.*
325

326 **Response:** Thank you for your insightful question regarding the methodological com-
327 parison. We conducted a performance comparison of named entity recognition (NER)
328 between GPT-4 and Spert.pl [8], a representative traditional NER approach, based on
329 the following key considerations:

- 330 • **Architecture Dependency:** Spert.pl fine-tunes a pre-trained BERT model using
331 supervised learning on annotated data, whereas GPT-4 operates in a zero-shot
332 generative manner, eliminating the need for task-specific fine-tuning.
- 333 • **Task-Specific Design:** Spert.pl employs custom span processing and threshold mech-
334 anisms tailored specifically for NER, whereas GPT-4 leverages its general-purpose
335 text generation capabilities to recognize entities dynamically.
- 336 • **Closed Ontology Constraints:** Spert.pl requires a predefined set of entity and relation
337 types, limiting its adaptability. In contrast, GPT-4 can extend its recognition scope
338 dynamically through natural language prompts, offering greater flexibility.

339 For a detailed comparison of GPT-4 and Spert.pl on biomedical entity recognition, we
340 have included the performance evaluation in the appendix.

341 **[Change made in Appendix D.5]**
342

343 **Comparative Performance of GPT-4 and Spert.pl in Named Entity Recog-** 344 **nition Task**

345 We evaluated the performance of Spert.pl (a BERT-based model) and GPT-4 on
346 a named entity recognition (NER) task. Spert.pl was fine-tuned on 400 manually
347 annotated biomedical abstracts for NER and relationship extraction tasks, while GPT-
348 4 was assessed in a few-shot setting. Benchmarking was conducted on a held-out test
349 set of 100 annotated abstracts distinct from Spert.pl’s training data. The F1 scores
350 across entity types are summarized in Table 4.

- 351 • **Performance on Well-Defined Entities:** GPT-4 demonstrated superior accuracy
352 (F1 scores: 0.92–0.94) for entities with clear definitions, including Disease/Symptom,
353 Gene, Anatomy, and Drug/Chemical. This high performance is likely attributable
354 to its extensive pretraining on diverse corpora, including biomedical datasets.
- 355 • **Ambiguous Entity Challenges:** Spert.pl achieved parity or outperformed GPT-
356 4 on entities with ambiguous boundaries, such as Method/Procedure (F1: 0.90 vs.
357 0.84) and Health Factors (F1: 0.82 vs. 0.80), suggesting task-specific training data
358 enhances resolution of contextually complex entities.

Table 4 Comparison of F1 Scores for Spert.pl and GPT-4 on Named Entity Recognition

Entity Type	Spert.pl	GPT-4	Sample Size (n)
Disease/Symptom	0.89	0.94	478
Gene	0.83	0.92	191
Health Factors	0.82	0.80	435
Method/Procedure	0.90	0.84	836
Anatomy	0.82	0.93	170
Drug/Chemical	0.88	0.93	180
Phenotype	0.78	0.83	86
Physiology	0.86	0.89	279
Sign	0.80	0.83	68

Note: Spert.pl was fine-tuned on 400 abstracts; GPT-4 was evaluated Few-shot. Test data comprised 100 non-overlapping annotated abstracts.

- **Data Dependency of Traditional Models:** The performance variability of Spert.pl (F1: 0.78–0.90) highlights the impact of training data quality and diversity, especially for low-sample entities such as Phenotype and Sign.
- **Practical Considerations:** Task requirements and cost-efficiency trade-offs should guide model selection.

10. For the MDD study, how is the 2000 individual control chosen? Also, why choose MDD? This procedure seems to be standardized across other mental health disorders as well - to show robustness, maybe the authors could experiment on other diseases as well.

Response: Thank you for your insightful question and valuable suggestions. The control individuals were selected based on the absence of any recorded diagnosis of psychological disorders. Specifically, we excluded individuals with any of the Phecodes ranging from 295 to 306.99 (see Table 5), which encompass various psychological conditions. To ensure a balanced and statistically robust comparison, we adopted a 1:2 ratio of MDD patients to controls. A comprehensive list of excluded Phecodes is provided in the Appendix E.

Response for "Why MDD": We selected MDD for our study because it is one of the most prevalent mental health disorders, characterized by well-established diagnostic criteria and a substantial impact on public health.

Response for "Experiment on other diseases": Based on your suggestion, we expanded our analysis to include anxiety and bipolar disorders, two prevalent and clinically significant mental health conditions. We applied the same experimental design—non-medical EHR factors only, medical embeddings only, and a combination of both—to these disorders. The performance of four models (SVM, LR, RF, and XGBoost), evaluated using AUC, is now presented in the updated Figure 4d. The results confirm that our approach is not only effective for MDD but also generalizable to other mental health conditions, further demonstrating the robustness of our method across various psychiatric disorders.

To enhance clarity, we have added the list of excluded Phecodes for the control group to the Appendix E. Additionally, we have revised the section on mental health disease prediction to provide a more detailed explanation of the experimental design and results.

[Change made in Main Text 2.5 Enhancing MDD Prediction through Knowledge Graph Embeddings]

We focused on a case group of patients diagnosed with MDD, anxiety, and bipolar disorder, compiling their medical records from 1 to 7 years prior to their first recorded diagnosis to construct pre-diagnosis health profiles. To enhance reliability, we excluded individuals with only a single recorded diagnosis, as these may represent potential misdiagnoses. For the control group, we selected individuals with no recorded diagnosis of psychological disorders, specifically excluding those with PheCodes ranging from 295 to 306.99 (see Table 5) to eliminate cases with any psychological conditions. We adopted a 1:2 case-to-control ratio, ensuring a balanced and statistically robust comparison. The baseline demographic and personality characteristics of participants are detailed in Table 7.

[Change made in Main Text 2.5.1 Integration of Knowledge Graph Embeddings Enhances Predictive Accuracy for Mental Health Disorders]

Our study presents a novel approach to representing a patient’s medical history using KG embeddings (see Figure 4g and Online Methods 4.4). Traditional models often encode diseases as dummy variables, which can lead to the curse of dimensionality, particularly in studies with limited sample sizes. For instance, representing 1,578 phenotypes typically requires an equally high-dimensional variable space, yet the available training samples may be insufficient to support such complexity. To address this, we introduce a KG embedding model that condenses a patient’s medical history into a 200-dimensional feature vector. This representation not only reduces the model’s dimensional burden but also integrates medical background knowledge from the knowledge graph, potentially improving diagnostic predictions and clinical insights.

Our analysis demonstrates that integrating medical KG embeddings with EHR-based factors significantly enhances prediction accuracy for MDD, bipolar disorder, and anxiety. Using ten-fold cross-validation across four classifiers (logistic regression [LR], random forest [RF], support vector machine [SVM], and XGBoost), we systematically evaluated three scenarios: (1) KG embeddings only, (2) non-medical EHR factors only, and (3) their integration. Performance was assessed using area under the ROC curve (AUC) with standard deviation, as shown in Figure 5.

- **Major Depressive Disorder:** When utilizing comprehensive non-medical EHR factors (see Supplementary for details), the integrated model (EHR + KG) achieved superior performance. XGBoost attained the highest AUC of 0.89 ± 0.02 , improving by 0.07 over the EHR-only model (0.82 ± 0.03) and by 0.18 over the KG-only model (0.71 ± 0.02). Similar trends emerged for LR (The integrated model: 0.88 ± 0.03 vs. EHR-only: 0.84 ± 0.04 ; KG-only: 0.75 ± 0.02) and RF (The integrated model:

0.86 $\pm$ 0.02 vs. EHR-only: 0.82 $\pm$ 0.03). Notably, models relying solely on environmental EHR factors exhibited a significant performance boost when enhanced with KG embeddings. XGBoost improved from 0.63 $\pm$ 0.03 (EHR-only) to 0.75 $\pm$ 0.02, reflecting a gain of 0.12. Similarly, SVM's EHR-only performance (0.57 $\pm$ 0.06) was relatively weak but improved significantly to 0.78 $\pm$ 0.03 after incorporating KG embeddings.

- **Bipolar Disorder:** XGBoost achieved an AUC of 0.88 $\pm$ 0.02, outperforming EHR-only (0.85 $\pm$ 0.02) and KG-only (0.66 $\pm$ 0.03) approaches. LR demonstrated the highest relative improvement (+0.17), increasing from 0.70 $\pm$ 0.03 (KG-only) to 0.87 $\pm$ 0.02 (integrated model), highlighting the benefits of multimodal integration.
- **Anxiety Disorders:** The integrated model consistently improved AUC across all classifiers (range: 0.81–0.83), compared to KG-only (0.65–0.69) and EHR-only (0.78–0.79). XGBoost performed best (AUC: 0.81 $\pm$ 0.01 vs. 0.78 $\pm$ 0.01 for EHR-only). The non-medical EHR-only model for anxiety performed the weakest among the three conditions, suggesting that lifestyle, environmental, and familial factors may contribute differently to anxiety disorders compared to MDD and bipolar disorder.

Additionally, our findings indicate that integrating medical KG embeddings helps prevent performance degradation in feature-limited scenarios, such as when using only environmental factors for MDD.

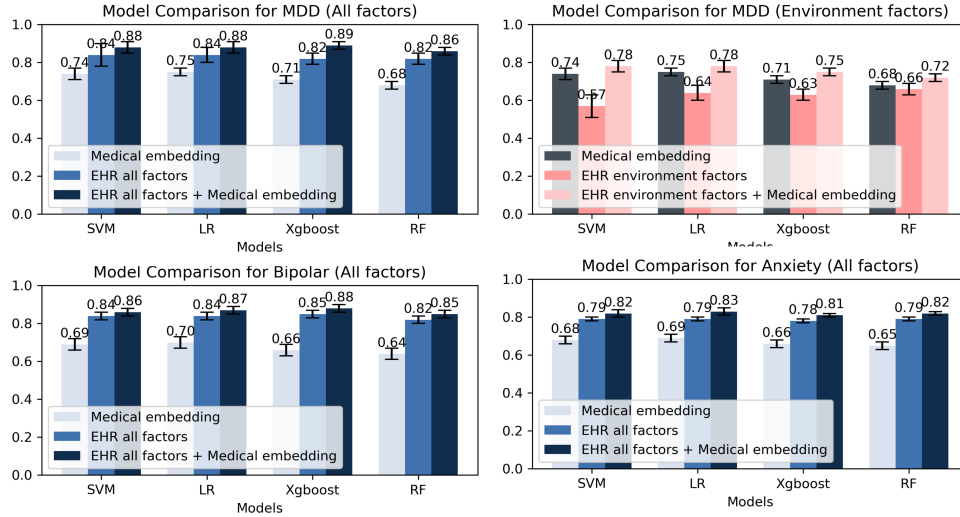


Fig. 5 Comparison of prediction performance across different diseases and models.

[Change made in Appendix E]

Table 5 Phecodes 295-306.99

Phecode	Phenotype
295.3	Psychosis
296	Mood disorders
300.1	Anxiety disorder
295.1	Schizophrenia
296.22	Major depressive disorder
301.1	Schizoid personality disorder
296.1	Bipolar
301	Personality disorders
300.4	Dysthymic disorder
300.13	Phobia
300.12	Agoraphobia, social phobia, and panic disorder
300.11	Generalized anxiety disorder
300.3	Obsessive-compulsive disorders
300.8	Acute reaction to stress
300.9	Posttraumatic stress disorder
304	Adjustment reaction
303.1	Dissociative disorder
303	Psychogenic and somatoform disorders
295	Schizophrenia and other psychotic disorders
303.4	Somatoform disorder
303.3	Psychogenic disorder
300	Anxiety, phobic and dissociative disorders
306	Other mental disorder
305.2	Eating disorder
305.21	Anorexia nervosa
302	Sexual and gender identity disorders
301.2	Antisocial/borderline personality disorder
306.9	Tension headache
297.2	Suicide or self-inflicted injury
297	Suicidal ideation or attempt

Table 6 The Excluded Phecodes for the Control Group

Characteristics	Anxiety		MDD		Bipolar	
	Control	Case	Control	Case	Control	Case
Sample size	2000	1000	2000	983	2000	1211
Age (years), mean (SD)	56.46 (8.16)	55.73 (8.46)	56.12 (8.09)	55.04 (8.43)	56.27 (8.27)	55.56 (8.34)
Female (%)	40.50	31.60	43.60	38.86	43.35	41.87
White (%)	90.85	91.10	89.65	88.20	88.25	88.11
Townsend index, mean (SD)	-1.56 (2.99)	-0.97 (3.21)	-1.53 (3.00)	-0.28 (3.50)	-1.49 (3.09)	-0.00 (3.53)
Employment status (%)	60.50	55.10	61.35	54.12	60.25	46.16
BMI, mean (SD)	27.02 (4.85)	27.58 (5.36)	26.98 (4.82)	27.98 (5.51)	26.87 (4.74)	28.30 (5.85)
Income categories (%)						
≤18,000	23.65	34.30	22.20	38.05	23.90	44.59
18,000-30,999	25.90	26.60	24.70	25.74	26.30	24.86
31,000-51,999	25.10	21.40	26.90	22.08	23.75	17.92
52,000-100,000	20.35	14.00	21.35	12.61	20.45	9.33
≥100,000	5.00	3.70	4.85	1.53	5.60	3.30

Table 7 Baseline characteristics of study participants stratified by disease status

452 11. *I feel like the approach to represent patient could be explained more clearly. Why*
453 *SapBERT? Why a simple aggregation of frozen KG embeddings? Can it learn end-to-*
454 *end using graph neural networks?*

455 **Response for "Why SapBERT":** We sincerely appreciate the reviewer's thoughtful
456 question and apologize for any lack of clarity in our initial explanation. In our disease
457 prediction framework, SapBERT is employed to compute semantic similarity between
458 medical entities, while RDF2vec is specifically used for generating medical embeddings
459 for patient representation. Below, we clarify the roles of these models in our approach:

- 460 • **SapBERT for Mapping ICD-10 Codes, Phecodes, and MDKG Entities.**
461 We utilize SapBERT to map entities from the MDKG to ICD-10 codes and Phe-
462 codes. SapBERT is particularly well-suited for this task because it enables the
463 computation of text embeddings for disease entities and ICD-10 codes, effectively
464 capturing their semantic similarity. Since SapBERT is pre-trained on the Unified
465 Medical Language System (UMLS)—which contains a rich set of biomedical and
466 clinical terms—it provides a robust understanding of the semantic relationships
467 between medical concepts. By leveraging SapBERT's capability to capture these
468 relationships, we can accurately align MDKG entities with their corresponding ICD-
469 10 codes and Phecodes, facilitating the construction of a knowledge graph grounded
470 in standardized medical classifications.
- 471 • **Graph Embedding using RDF2vec.** Once we have mapped the MDKG entities
472 to Phecodes, we generate graph embeddings using RDF2vec. This model trans-
473 forms the structural relationships within the knowledge graph into fixed-dimensional
474 vector representations by applying random walks over the graph. This process effec-
475 tively captures the connections between Phecodes and other medical entities in the
476 MDKG. RDF2vec is particularly advantageous because it efficiently extracts mean-
477 ingful relationships from graph data without incurring the high computational costs
478 associated with more complex models, such as Graph Neural Networks (GNNs).

479 By employing SapBERT for entity mapping and RDF2vec for graph-based representa-
480 tion, we ensure a semantically rich and computationally efficient approach to medical
481 entity embedding and disease prediction.

482 **Response for "Why a simple aggregation of frozen KG embeddings":** Thank
483 you for your insightful comments. Our decision to adopt a simple mean aggregation
484 of frozen KG embeddings is guided by two key considerations:

- 485 • **Simplicity and Efficiency:** Mean aggregation offers a computationally efficient
486 and straightforward method for integrating information across multiple nodes[9,
487 10]. By averaging pre-trained embeddings, we eliminate the need for additional
488 parameters or complex computations, ensuring minimal computational overhead.
489 Since each node inherently captures its own semantic properties, their averaged
490 embeddings provide a representative central tendency in the semantic space.
- 491 • **Semantic Consistency Assumption:** Medical history entities typically belong to
492 the same semantic hierarchy (e.g., different phenotypes), meaning their embeddings

493 share common features. By leveraging mean aggregation, we preserve these shared
494 characteristics, making it a logical and effective choice for embedding integration.

495 This approach ensures both efficiency and semantic coherence while avoiding unnec-
496 essary computational complexity.

497 **Response for "Can it learn end-to-end using graph neural networks":** We
498 did not incorporate graph-based models (such as GNNs) in this study due to their
499 high computational demands and challenges in training on large-scale graphs like ours.
500 Instead, we opted for RDF2vec, which provides a simpler and more efficient approach
501 while still capturing essential graph structural information.

502 Additionally, we have revised the section on patient representation and updated the
503 corresponding pipeline figure to enhance clarity and improve the overall structure of
504 the presentation. We hope these refinements make our approach more comprehensible.

505 **[Change made in Online Methods 4.4 Representing Patient’s Medical His-**
506 **tory using Knowledge Graph Embedding]**

508 After collecting a patient’s medical history, we generate their medical knowledge graph
509 embeddings through the following steps:

- 510 1. Our approach begins by mapping entities from MDKG to ICD-10 codes by calculat-
511 ing the SapBERT embeddings for each entity and ICD-10 codes, then determining
512 their similarity. Using the established mapping between ICD-10 codes and Phe-
513 codes, we further map the MDKG entities to their corresponding Phecodes.
514 SapBERT is particularly well-suited for this task due to its pre-training on the
515 UMLS, a comprehensive database that includes a wide array of biomedical and clin-
516 ical terms. This pre-training enables SapBERT to effectively capture the semantic
517 relationships between medical concepts, allowing for accurate mappings of MDKG
518 entities to Phecodes.
- 519 2. After mapping MDKG entities to Phecodes, we compute graph embeddings for
520 those entities that successfully map to Phecodes using the RDF2vec^[11] model.
521 RDF2vec is a highly efficient technique that converts the structure of the knowledge
522 graph into fixed-dimensional vector representations. Compared to Graph Neural
523 Networks (GNNs), RDF2vec requires far fewer computational resources, making it
524 a more feasible option for large-scale applications like ours.
- 525 3. Next, we average the graph embeddings of the entities that map to the same
526 Phecode to generate the final Phecode embedding.
- 527 4. To generate the final patient embedding, we average the embeddings of all Phecodes
528 recorded for the patient in the UK Biobank. This process results in a compact
529 and informative 200-dimensional medical embedding that summarizes the patient’s
530 medical history for diagnostic predictions.

531 *12. Why is air pollution an important factor for MDD?*

532 **Response:** Thank you for your insightful comment.

533 To clarify, the SHAP plot in Figure 4e presents the results of two models: one using
534 only environmental factors and another incorporating both environmental factors and

535 medical embeddings. As shown in the plot, the model based solely on environmental
 536 factors achieves the lowest AUC, whereas integrating medical embeddings leads to the
 537 most significant improvement. The primary purpose of the SHAP plot is to examine
 538 the key factors driving this change.

539 Regarding the impact of air pollution on MDD, we conducted a comprehensive
 540 literature review using our knowledge graph. Initially, we identified five studies that
 541 directly support the hypothesis that air pollution is a risk factor for MDD (see Table
 542 8). To further strengthen our analysis, we expanded our search criteria to include
 543 studies examining the broader relationship between air pollution and depression. This
 544 extended search identified an additional 56 studies, all of which support the link
 545 between air pollution and depression.

546 Collectively, these findings suggest that air pollution, particularly exposure to fine
 547 particulate matter (PM2.5), plays a significant role in the onset or progression of
 548 MDD. This aligns with our model’s results, which highlight air pollution as a key
 549 environmental factor influencing MDD.

Table 8 How Air Pollution Affects MDD

Reason	PMID
Long-term exposure to air pollution increases the risk of schizophrenia, depression, and anxiety disorders: This large longitudinal cohort study highlights how long-term exposure to various air pollutants (PM2.5, NO, BC, UFP) increases the incidence of major depressive disorder (MDD), particularly through changes in brain networks that affect cognition and emotional health[?].	37944432
Real-time air pollution worsens depression symptoms: This study finds that localized air pollution worsens depressive symptoms in people with bipolar disorder. Though it didn’t find an association with mania symptoms, worsening depression symptoms with worsening air quality suggests a potential similar impact on MDD[12].	37313799
Genetic risk and air pollution affect brain networks involved in depression: This study explores how air pollution (PM2.5) interacts with genetic risk factors to impair brain functions associated with depression. Exposure to PM2.5 affects reasoning, working memory, and effective connectivity across brain networks implicated in depression[13].	34750260
Long-term exposure to air pollution is a risk factor for depression in China: This population-based cohort study found that exposure to pollutants like PM2.5, PM10, and NO was linked to a higher risk of developing depression over time, especially among individuals who did not take protective measures[?].	34798713
Chronic ozone exposure leads to depressive symptoms and oxidative stress: In this rodent model, ozone exposure resulted in memory impairment, anxiety, and depression-like behavior. The exposure induced oxidative stress, similar to what is observed in depression, and diminished the effectiveness of antidepressants[14].	25877744

550 We have updated the manuscript to reflect this clarification by modifying the
 551 original text:

552 **[Change made in Main Text 2.5.2 Understanding Feature Importance with**
 553 **SHAP Values]**

554

555 In Scenario 1, which considers only environmental factors, PM2.5 emerged as the most
556 influential feature with a SHAP value of 0.43. This highlights the significant role of air
557 pollution, particularly fine particulate matter, in MDD risk. This finding aligns with
558 previous studies that link air pollution to depressive symptoms through mechanisms
559 such as changes in brain networks, which may affect cognition, memory, and emotional
560 health [13, 15]

561 *13. Why is non-medical EHR embedding have such a good performance? Is there*
562 *potential confounders? How do you select these variables to avoid confounders?*

563 **Response:** Thank you for your insightful question. We selected non-medical variables
564 related to MDD by querying our MDKG and expanded our selection within each
565 category using data from the UK Biobank. The selected variables cover a wide range
566 of domains, including demographics, lifestyle, environmental exposures, and health
567 behaviors. Examples include:

- 568 • **Demographics:** Age, sex, ethnicity, socioeconomic status (e.g., household income,
569 Townsend deprivation index).
- 570 • **Lifestyle:** Exercise frequency, dietary habits (e.g., fruit intake, processed meat
571 intake), sleep duration, and TV watching time.
- 572 • **Environmental Factors:** Air pollution (e.g., PM2.5, nitrogen oxides), greenspace
573 percentage, and noise pollution.
- 574 • **Health Behaviors:** Smoking status, alcohol intake, and medication use.
- 575 • **Family History:** Illnesses of parents and siblings.

576 A complete list of variables is provided in the manuscript (Supplementary).

577 Addressing Potential Confounders

578 To minimize the impact of confounders, we took the following two steps:

- 579 • **Multivariate Models:** We used multivariate models such as logistic regression,
580 SVM, XGBoost and random forests to control for the effect of multiple variables
581 simultaneously. This allows us to account for the impact of confounding factors like
582 age, sex, socioeconomic status, and family history, ensuring that these factors do
583 not influence the relationships observed.
- 584 • **Cross-validation:** We employed 10-fold cross-validation to ensure that the mod-
585 els were not overfitting and that the results were robust. This helps to assess the
586 generalizability of the model across different subsets of the data.

587 *14. How do the authors plan to visualize the KG and deliver to the users? Currently,*
588 *the github is not available.*

589 **Response:** Thank you for your comment. We have created a webpage for querying
590 and downloading relevant data from the MDKG: <https://biomedkg.com>, and we are
591 continuously working on improving it. Additionally, we have updated our GitHub
592 repository to enhance accessibility and usability of the code.

Entity I Name

Entity I Type

Select

Entity II Name (optional)

Entity II Type (optional)

Select

Relation Type (optional)

Select

Search

The image displays the BiomedKG website interface. On the left is a search form with fields for 'Entity I Name', 'Entity I Type' (a dropdown menu), 'Entity II Name (optional)', 'Entity II Type (optional)' (a dropdown menu), 'Relation Type (optional)' (a dropdown menu), and a 'Search' button. On the right is a detailed entity profile for ID:0000002. The profile includes a circular diagram showing relationships between 'Adverse Drug Reaction', 'Drug', 'Disease', 'Symptom', 'Phenotype', 'Gene/Protein', 'Disease/Symptom', 'Drug/Chemical', and 'Phenotype'. The profile also contains a table of 'Contextual Information' with columns for 'Entity Types', 'Relationship Types', and 'Contextual Information'. The table lists various entity types such as Health Factors, Anatomy, Phenotype, Sign, Method/Procedure, Physiology, Gene/Protein, Disease/Symptom, Drug/Chemical, and various relationship types like Risk, Diagnostic, Characteristic, Association, Therapeutic, Pharmacological, and Abbreviation.

Fig. 6 BiomedKG website

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Response for Reviewer 2

1. Use of ChatGPT: While ChatGPT demonstrates impressive performance across many benchmarks, the choice of a closed, proprietary platform for constructing a shared knowledge graph is problematic. A key expectation for a knowledge graph project is that it should be built on an openly available large language model (LLM), rather than a closed system like OpenAI's platform. The use of closed systems hinders transparency in understanding the deeper structures of the knowledge graph, raising concerns about the potential for data misuse by third-party providers.

Response: We appreciate the reviewer's concern regarding the use of OpenAI's ChatGPT for constructing the knowledge graph. The decision to use OpenAI's API was based on three key factors:

- **Model capability.** OpenAI's API offers state-of-the-art language models that perform well in tasks related to complex domain-specific knowledge, such as relationship extraction.
- **Response speed.** OpenAI's models provide quick responses, which is critical for the timely processing of large datasets.
- **Minimal computational resource requirements.** Our approach does not require high computational resources, which makes OpenAI's API a good fit for our framework, as it provides efficient performance without the need for intensive processing power.

However, it is important to emphasize that our work presents a flexible framework for knowledge graph construction, where each component can be replaced with more advanced or open-source models. The framework is not dependent on OpenAI's platform. For instance, if alternative models such as DeepSeek, LLaMA, or other open-source solutions were used, the framework would remain fully functional. This adaptability ensures that our approach can evolve over time, integrating the best available models while maintaining broad compatibility across different tools and platforms.

2. Reliability of Insights: There is a lack of clarity regarding how users can verify the reliability of insights derived from the knowledge graph, particularly those not covered by established medical databases. It is crucial when employing LLMs that users are encouraged to assess machine-generated content rather than blindly trust it critically. Hallucinations—a known issue in many domains involving LLMs—pose a specific risk. How has this been addressed or mitigated in the current implementation?

Response: Thank you for your valuable feedback regarding the reliability of insights derived from our knowledge graph, particularly for information not covered by established medical databases. We fully recognize the importance of ensuring that users can assess the credibility of machine-generated content, especially when utilizing LLMs. To address this concern, we have implemented several measures to enhance the reliability of our knowledge graph:

- **Technical Reliability of the Extraction Process:** To ensure accurate extraction of entities and relationships from the literature, we employ Spert.pl, a model that has demonstrated strong performance in entity recognition (F1 score: 0.89) and relationship extraction (F1 score: 0.68) on a validation set. While these results indicate high reliability, we further employ a multi-step refinement process for the extracted triplets. The GPT-4o is used to refine these triplets through a structured validation process that includes: (i) Verifying whether entity spans are correctly identified. (ii) Ensuring that the entities are of the correct type. (iii) Validating the relationship between the identified entities. and (iv) Assigning a confidence score (1-10) to assess the overall accuracy of the triplet. Triplets scoring below 5 undergo further refinement to enhance their accuracy. This iterative approach helps ensure that the extracted knowledge is more reliable and meaningful. Following the post-processing refinement steps, significant improvements are observed, with the joint model’s micro F1 score rising to 0.79 and the macro F1 score increasing to 0.76, demonstrating a marked enhancement in overall performance.
- **Data Source Reliability:** The reliability of insights also depends on the quality of data sources from which triplets are derived. To assess and communicate this reliability, we track two key factors:
 - Frequency Across Sources:** We monitor how often each triplet appears in multiple sources, including scientific literature and integrated databases. A higher frequency across diverse sources suggests greater reliability.
 - Confidence Score:** We assign a confidence score to each literature-derived triplet, incorporating multiple factors: model prediction scores, journal impact factor, frequency across sources, number of distinct publication years, most recent publication year, and contradictions within the data[1–3]. This confidence score provides a quantitative measure of reliability, enabling users to assess the trustworthiness of each triplet. Higher confidence scores indicate greater reliability, whereas lower scores suggest that further evaluation may be needed. Additionally, the manual evaluation revealed a strong positive correlation (mean correlation: 0.78) between our confidence score and human-assigned evaluation scores, further validating its effectiveness.

Based on these improvements, we have revised the relevant section of the manuscript to ensure a clearer and more structured presentation of our reliability framework.

[Change Made in Online Methods 4.2.1 Named Entities, Relationships, and Conditional Statements Extraction]

After extracting entities and relationships with the BERT-based model, we refined the extracted triplets using GPT-4o. The refinement process includes verifying entity

spans, confirming entity types, validating relationships, and scoring each triplet (1-10) for overall accuracy. Triplets with a score below 5 are reprocessed. In cases of contradictory statements, we input the conflicting triplets along with the full abstract into GPT-4o to determine whether the contradictions stem from differing conclusions in the articles. If the contradictions are due to differing conclusions, they are retained; otherwise, the model selects the correct triplet, removing the incorrect one from the knowledge graph.

The best-performing model demonstrates high accuracy in entity recognition, achieving a micro F1 score of 0.89 and a macro F1 score of 0.88, indicating reliable precision across different entity types. Relationship extraction presents more challenges, with initial micro and macro F1 scores of 0.68 and 0.67, respectively. When both tasks are performed jointly, the model achieves micro and macro F1 scores of 0.63 and 0.63, highlighting the complexity of integrating these processes. However, following the post-processing refinement steps, significant improvements are observed, with the joint model’s micro F1 score rising to 0.79 and the macro F1 score increasing to 0.76, demonstrating a marked enhancement in overall performance.

[Change Made in Main Text 2.1.2 Incorporation of Contextual Information]

To assess the credibility of triplets within MDKG, we incorporate multiple factors that influence their confidence levels. Reliability is first evaluated through cross-source consensus, where triplets appearing in multiple sources are considered more trustworthy. For literature-derived triplets, we further document key attributes such as publication type (e.g., reviews, clinical trials, meta-analyses) and analytical methods used (e.g., multivariate analysis, multiple regression, logistic regression), as these factors contribute to varying levels of evidentiary strength. To enhance reliability assessment, we assign a confidence score (detailed in Appendix C) to each triplet. This score integrates multiple factors, including model prediction scores, journal impact, frequency across sources, distinct publication years count, maximum publication year, and contradiction counts in the data. By appending this confidence score to each triplet, MDKG enables users to prioritize high-quality data and better assess the reliability of the extracted knowledge.

[Change Made in Appendix C]

Confidence Score Calculation

The confidence score integrates multiple quality indicators of scientific statements (triples) using a weighted framework. Each component score is normalized to [0,1] and scaled by a predefined weight, ensuring the final score remains within the range [0,3]. Prior research on assessing study quality has highlighted key factors influencing a study’s credibility and impact, including impact factor, statement frequency, and the number of distinct publication years^[1-3]. Our framework incorporates these established factors and further refines their contributions based on insights from domain experts, ensuring a balanced and context-aware evaluation. The confidence score is computed

119 as follows:

$$S_{confidence} = S_{prediction} + 0.8 \cdot S_{IF} + 0.6 \cdot S_{Frequency} + 0.4 \cdot S_{years} + 0.2 \cdot S_{date} - 0.5 \cdot S_{contradiction}$$

- 120 • **Model Prediction Score** ($S_{prediction}$): Quantifies the base trustworthiness of the
 121 triple generated by the prediction model. The model directly outputs the score,
 122 ranging from $[0, 1]$.
- **Impact Factor Score** (S_{IF}): Reflects the academic influence of the resource.

$$S_{IF}(i) = 1 - \exp(-\alpha i)$$

123 where i is the average impact factor of the source journal. To align with the 75th
 124 percentile impact factor value (6.6), the decay coefficient $\alpha = 0.25$ ensures $S_{IF} \geq 0.8$
 125 when $i \geq 6.6$.

- **Frequency Score** ($S_{Frequency}$): Measures how frequently the triple appears across different resources.

$$S_{IF}(n) = 1 - \exp(-\alpha(n-1))$$

126 where n is the occurrence count of the triple. With the 75th percentile $n = 4$, $\alpha =$
 127 0.55 ensures $S_{Frequency} \geq 0.8$ when $n \geq 4$.

- **Distinct Publication Years Counts Score** (S_{years}): Captures the historical consistency of the triple across years.

$$S_{years}(n_t) = 1 - \exp(-\alpha n_t)$$

128 where n_t is the number of distinct publication years. For the 75th percentile $n_t =$
 129 4 , $\alpha = 0.4$ yields $S_{years} \geq 0.8$ when $n_t \geq 4$.

- **Max Publication Year Score** (S_{date}): Rewards recency of the triple's latest publication.

$$\begin{cases} S_{date}(t) = \exp(-\alpha(T-t)), & t \leq T \\ S_{date}(t) = 1, & t > T \end{cases}$$

130 where T is the reference year (e.g., 2025) and t is the latest publication year. The
 131 Decay Rate $\alpha = 0.1$ imposes gradual penalties for older publications.

- **Contradiction Counts Score** ($S_{contradiction}$): Penalizes conflicting statements from other sources.

$$S_{contradiction}(c) = \exp(-\alpha c)$$

132 where c is the count of contradictory triples. $\alpha = 0.1$ ensures a penalty of 0.8 when
 133 $c = 2$.

134 In summary, the confidence score aggregates multiple aspects of a resource's qual-
 135 ity and relevance, with each factor having a distinct role in shaping the final score.
 136 The decay functions ensure that the score is adjusted for diminishing returns, while
 137 the weights reflect the relative importance of each factor in the overall assessment.

138 This approach allows for a nuanced evaluation of resources, balancing credibility, aca-
 139 demic impact, historical significance, recency, and frequency of use. The distribution
 140 of confidence scores across different types of relationships is illustrated in Figure 1.

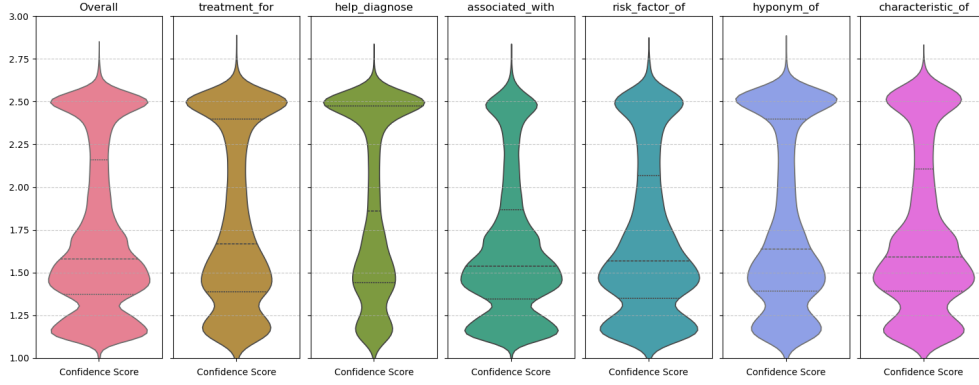


Fig. 1 The distribution of confidence scores for different relationships.

141 3. Lack of Comparable Improvements: While the authors claim that the inclusion of
 142 "side facts" in their knowledge graph is a key advantage, the report does not provide
 143 sufficient details for comparing this system with existing knowledge graphs. Moreover,
 144 the heavy reliance on literature (over 90%) in constructing the graph casts doubt on the
 145 robustness of their approach, particularly with respect to the integration of databases.

146 **Response for 'side facts' in our KG:** Thank you for raising this important point.
 147 Upon reviewing recent knowledge graphs, we found that most do not include side
 148 facts[4, 5], and only a few incorporate common contextual features[6–8] (e.g., those
 149 highlighted in black in Table 1). However, for more complex contextual information
 150 that requires extraction from text (shown in blue in Table 1), current knowledge graphs
 151 do not appear to address these aspects, making direct comparisons challenging.

152 While direct comparisons may be difficult, we have included examples in the
 153 manuscript that demonstrate the use of these side facts. For instance, in the Group-
 154 Specific Mental Health Concerns section, we identified significant differences in mental
 155 health conditions between adolescent and elderly populations. This was enabled by
 156 extracting the "occurs in" contextual feature, which indicates the population group
 157 where a relationship is relevant (if mentioned in the text). Additionally, we visual-
 158 ized these findings using a word cloud, showcasing the populations most frequently
 159 referenced in mental health research and highlighting the primary focus of existing
 160 studies.

161 These contextual features enhance the reliability of extracted triplets by incorpo-
 162 rating the quality and context of the information. Furthermore, the confidence score

provides users with a quantitative measure of reliability, allowing for more informed interpretation and application of the extracted knowledge.

Category	Contextual Features
Conditional Statements Baseline Characteristics	occurs in, located in Average age, Gender distribution, Racial composition, Educational attainment, Employment status, Overall sample size, and Cohort name
Source Information	Source, Authors, Year, Journal, DOI, Pmid, sentence, impact
Confidence	Confidence score, frequency, statistical analysis method, publication type

Table 1 Contextual Features for MDKG

Response for the heavy reliance on literature: Thank you for your comment regarding the reliance on literature and its potential impact on the robustness of our approach. While our knowledge graph incorporates literature-based insights, it is important to clarify that literature-derived triplets constitute only about 20% of the final MDKG, while the remaining 80% is sourced from trusted, well-established biomedical databases. This integration significantly enhances the overall robustness and reliability of the knowledge graph.

During the data integration process, we carefully evaluate the quality of biomedical databases based on key factors such as data updates and manual curation. We ensure that the databases are regularly updated to reflect the latest research and medical advancements. Additionally, preference is given to databases that have undergone manual expert review, ensuring greater accuracy and trustworthiness of the integrated data.

The literature-based triplets are extracted from peer-reviewed research articles, meaning they have already undergone rigorous academic validation. To further enhance insight reliability, we assign a confidence score to each extracted triplet. This scoring system allows users to place greater trust in high-confidence triplets, as they are expected to be more robust, while critically assessing lower-confidence triplets with appropriate caution.

In summary, the robustness of our approach is reinforced through the careful integration of literature-derived triplets with high-quality biomedical databases, along with the confidence scoring system, which provides users with a quantitative measure of reliability for the generated insights.

4. Superficial Examples: The examples provided in the study remain superficial, with many of the associations being either obvious or, in some cases, misinterpreted. For instance, an association between two entities does not automatically imply risk or causality. The examples presented, such as the one involving anxiety, could easily be found by any researcher conducting a PubMed search, which raises the question of what unique benefit this knowledge graph offers.

Response for an association between two entities does not automatically imply risk or causality: Thank you for raising this important issue. As you correctly pointed out, correlation does not imply causation. In medical research, unless explicitly using causal inference models (e.g., structural equation models or Mendelian randomization), conclusions drawn from standard regression analyses (e.g., logistic regression, Cox models, linear regression) often cannot establish causality directly. While directional terms such as "leads to" are frequently used in the literature, they do not necessarily imply risk or causation.

Despite this limitation, directional keywords remain essential in our model's classification of "risk factor" relationships. Our knowledge graph extracts relationships at the sentence level, meaning the classification is largely influenced by the phrasing used in the original text. If a study presents a clear directional relationship, our model is likely to classify it as a "risk". However, at the report level (language level), mitigating this issue entirely is challenging, as we do not always have access to the underlying data and statistical methods used in the original study.

To address this limitation, we incorporate "statistical method" as a contextual feature. Each triplet extracted from the literature is assigned a "statistical method" tag, indicating the type of analysis supporting the conclusion in the source document. For instance, relationships identified using causal models such as structural equation models or Mendelian randomization are classified as high-confidence risk relationships. This contextual feature allows users to critically assess the reliability of these relationships and make more informed judgments based on the statistical approach used in the study.

Response for what unique benefit this knowledge graph offers compared with a PubMed search: Compared to a traditional PubMed search, our knowledge graph offers several key advantages:

- **Quick Access to Key Information:** Instead of manually reading through numerous papers, users can efficiently navigate a structured summary of core findings extracted from the literature, significantly reducing the time required to locate relevant information.
- **Uncovering Indirectly Connected Relationships:** Our knowledge graph not only visualizes direct relationships but also reveals potential associations through intermediate nodes. This feature helps users discover hidden connections and formulate new hypotheses based on previously unnoticed knowledge networks.
- **Integrated Multi-Source View:** Unlike PubMed searches that focus solely on literature, our knowledge graph integrates information from high-quality biomedical databases, including specialized datasets on genes, diseases, and drugs. This allows users to access both literature-based insights and structured database knowledge in a single platform, eliminating the need to query multiple sources separately.
- **New Insights Discovery:** Beyond organizing existing information, our knowledge graph leverages large-scale graph algorithms (e.g., link prediction) to identify previously unrecognized relationships, enabling the discovery of novel insights.
- **Precision Filtering:** To enhance usability, we embed contextual features into certain triplets, such as study population details or conditional statements. This enables

238 users to apply granular filtering, such as isolating findings specifically derived from
239 studies involving female cohorts. By offering refined search capabilities, our approach
240 allows researchers to conduct more targeted and precise explorations based on their
241 specific needs.

242 *5. Need for Thorough Analysis: The concept of an enriched knowledge graph is indeed*
243 *promising, but a more thorough analysis of the various stages involved is needed. The*
244 *rapid pace of developments in this field does not excuse the lack of rigorous validation*
245 *at each step of the process. This is particularly relevant when it comes to the UK*
246 *Biobank analysis, where the methodology for using the knowledge graph in predictions*
247 *remains entirely unclear. In fact, the scope of the current work seems too broad, and I*
248 *would recommend splitting the project into two separate works: one that focuses on the*
249 *in-depth analysis and validation of the knowledge graph itself, and another dedicated*
250 *to the UK Biobank project.*

251 **Response for validation at each step of the process:** Thank you for your insight-
252 ful suggestion. We have carefully reviewed and manually validated each step of the
253 knowledge graph construction process to ensure accuracy. For transparency, we have
254 included all results in the Appendix.

255 **[Change made in Appendix D]**

256 **Performance of Relationship Annotation with GPT-4**

257 We used GPT-4 for pre-annotation, significantly reducing the time and effort
258 required for manual annotation. GPT-4 performed two main tasks: (1) identifying noun
259 phrases in abstracts and (2) determining whether a specific relationship (e.g., thera-
260 peutic or risk) existed between annotated entity pairs. The second task was treated
261 as a binary classification problem.

262 We evaluated GPT-4’s performance by comparing its annotations with manually
263 reviewed data from 500 mental health-related abstracts (MDIEC). For entity identi-
264 fication, GPT-4 achieved an F1-score of 0.88, with a precision of 0.84 and a recall of
265 0.93, demonstrating strong recognition of relevant terms while occasionally including
266 non-essential ones. For relationship extraction, performance was assessed against the
267 manually reviewed dataset, with results summarized in Table 2.

268 These results indicate that GPT-4 achieves high performance in identifying well-
269 defined relationships (F1-scores above 0.9) but shows reduced accuracy for more
270 ambiguous categories, such as Characteristic, Risk, Therapeutic, and Occurs in. The
271 strong performance in relationship extraction is largely attributable to our reformula-
272 tion of the original multi-class task (where the model determined the most appropriate
273 relationship among nine possible types) into a binary classification task (where the
274 model was only asked whether a specific relationship existed). This reformulation
275 reduced categorical complexity, improving model accuracy, but significantly increased
276 the cost of model usage.

277 **Performance of Triplet Extraction with Spert.pl and Refinement with** 278 **GPT-4o**

Relationship Types	GPT4 F1 Score	Sample Size
Hierarchical	0.92	1166
Association	0.83	2686
Characteristic	0.81	700
Risk	0.85	1199
Therapeutic	0.83	1239
Diagnostic	0.91	589
Abbreviation	0.97	1112
Conditional Statement Types		
Occurs in	0.75	1225
Located in	0.90	524

Table 2 Relationship Types and Conditional Statement Types with GPT4 F1 Score and Sample Size

Triplet extraction was performed in two stages. First, a BERT-based classification model (Spert.pl) was fine-tuned on the MDIEC annotated dataset for named entity recognition (NER) and relationship extraction (RE) and then applied to unannotated abstracts to extract triplets. In the second stage, GPT-4o was used to validate and refine the extracted triplets, improving their accuracy and reliability.

1. The performance of Triplet Extraction

For the NER task, Spert.pl performed multi-class classification to identify and categorize potential entities. For the RE task, the model determined whether a specific relationship existed between a given pair of entities. As shown in Table , Spert.pl achieved strong performance in NER (macro F1-score: 0.88), but relationship extraction remained challenging (macro F1-score: 0.67). When evaluating the joint NER and RE performance—assessing whether both entities and their relationships were correctly identified—the model achieved a macro F1-score of 0.63.

2. The performance of Triplet Validation

To enhance the accuracy of extracted triplets, GPT-4o was employed for validation and refinement. After this process, the micro F1-score for triplet extraction improved to 0.79, with micro precision increasing to 0.84. This substantial improvement significantly enhanced the quality of the knowledge graph while maintaining cost efficiency.

Task	Micro-P	Micro-R	Micro-F1	Macro-P	Macro-R	Macro-F1
NER	0.89	0.90	0.89	0.88	0.88	0.88
RE	0.69	0.67	0.68	0.68	0.67	0.67
Joint NER & RE	0.62	0.65	0.63	0.62	0.65	0.63
Valid Joint NER & RE	0.84	0.74	0.79	0.83	0.70	0.76

Table 3 Performance metrics for NER, RE, and Joint NER & RE tasks

The performance of Table information extraction

We have conducted a manual review to assess the performance of extracting baseline characteristics from PDF tables. This review involved evaluating the accuracy of the process and identifying the causes of any errors encountered. The results of this evaluation are presented in Table 4. Out of the 302 total documents reviewed, 238 were

Table 4 Baseline Characteristics Extraction Error Analysis

Total number of documents	302
Total PDFs	238
PDFs with target tables	72
Error 1. Tables not found: 12	
Reason 1: Issues with table titles across pages	2
Reason 2: Issues with keywords	3
Other Reasons	7
Error 2. The table found but the final answer is wrong: 10	
Reason 1: Incomplete table	1
Reason 2: Header issues (headers not extracted)	3
Reason 3: Horizontal tables	1
Reason 4: Issues attributed to ChatGPT	5
a. Incorrect or incomplete grouping with multiple headers	
b. Inaccurate identification of categories such as education and employment	
c. Still provides answers when incorrect tables are input.	

PDFs, among which 72 contained target tables expected to be extracted for analysis.

The review identified two primary types of errors. The first, Tables Not Found, occurred in 12 documents. This issue was primarily caused by two factors: (1) in two instances, the system failed to detect tables because their titles spanned multiple pages, making it difficult to correctly associate the title with the corresponding data; (2) in three cases, the system struggled to identify tables due to unclear or missing keywords (in our method, we differentiate tables by locating baseline-related keywords). The remaining seven cases were attributed to less identifiable reasons, likely related to complex formatting or irregular data layouts.

The second error type, Table Found but Final Answer is Wrong, accounted for 10 documents. The primary causes included incomplete table extraction, missing table headers, issues with vertical table formats, or incorrect answers generated by ChatGPT.

Response for the UK Biobank analysis: Thank you for your comments. We have revised the manuscript to provide a clearer explanation of how we integrate knowledge graph embeddings (KG embeddings) for disease prediction. We hope these modifications enhance the clarity of our approach and improve the overall readability of the paper.

Furthermore, to demonstrate the robustness of integrating non-medical factors from electronic health records (EHR) with medical KG embeddings for mental disease prediction, we have extended our experimental setup. Specifically, we have expanded

the range of predictive models, increasing the number from two (Logistic Regression and SVM) to four by incorporating XGBoost and Random Forest. Additionally, we have broadened the scope of disease prediction beyond major depressive disorder (MDD) to also include anxiety and bipolar disorder. We believe these enhancements strengthen our findings and provide a more comprehensive evaluation of our approach.

[Change made in Online Methods Representing 4.4 Patient’s Medical History using Knowledge Graph Embedding]

After collecting a patient’s medical history, we generate their medical knowledge graph embeddings through the following steps:

1. Our approach begins by mapping entities from MDKG to ICD-10 codes by calculating the SapBERT embeddings for each entity and ICD-10 codes, then determining their similarity. Using the established mapping between ICD-10 codes and Phecodes, we further map the MDKG entities to their corresponding Phecodes. SapBERT is particularly well-suited for this task due to its pre-training on the UMLS, a comprehensive database that includes a wide array of biomedical and clinical terms. This pre-training enables SapBERT to effectively capture the semantic relationships between medical concepts, allowing for accurate mappings of MDKG entities to Phecodes.
2. After mapping MDKG entities to Phecodes, we compute graph embeddings for those entities that successfully map to Phecodes using the RDF2vec^[9] model. RDF2vec is a highly efficient technique that converts the structure of the knowledge graph into fixed-dimensional vector representations. Compared to Graph Neural Networks (GNNs), RDF2vec requires far fewer computational resources, making it a more feasible option for large-scale applications like ours.
3. Next, we average the graph embeddings of the entities that map to the same Phecode to generate the final Phecode embedding.
4. To generate the final patient embedding, we average the embeddings of all Phecodes recorded for the patient in the UK Biobank. This process results in a compact and informative 200-dimensional medical embedding that summarizes the patient’s medical history for diagnostic predictions.

[Change Made in Main Text 2.5.1 Integration of Knowledge Graph Embeddings Enhances Predictive Accuracy for Mental Health Disorders]

Our study presents a novel approach to representing a patient’s medical history using KG embeddings (see Figure 4g and Online Methods 4.4). Traditional models often encode diseases as dummy variables, which can lead to the curse of dimensionality, particularly in studies with limited sample sizes. For instance, representing 1,578 phenotypes typically requires an equally high-dimensional variable space, yet the available training samples may be insufficient to support such complexity. To address this, we introduce a KG embedding model that condenses a patient’s medical history into a 200-dimensional feature vector. This representation not only reduces the model’s dimensional burden but also integrates medical background knowledge from the knowledge graph, potentially improving diagnostic predictions and clinical insights.

Our analysis demonstrates that integrating medical KG embeddings with EHR-based factors significantly enhances prediction accuracy for MDD, bipolar disorder, and anxiety. Using ten-fold cross-validation across four classifiers (logistic regression [LR], random forest [RF], support vector machine [SVM], and XGBoost), we systematically evaluated three scenarios: (1) KG embeddings only, (2) non-medical EHR factors only, and (3) their integration. Performance was assessed using area under the ROC curve (AUC) with standard deviation, as shown in Figure 2.

- **Major Depressive Disorder:** When utilizing comprehensive non-medical EHR factors (see Supplementary for details), the integrated model (EHR + KG) achieved superior performance. XGBoost attained the highest AUC of 0.89 ± 0.02 , improving by 0.07 over the EHR-only model (0.82 ± 0.03) and by 0.18 over the KG-only model (0.71 ± 0.02). Similar trends emerged for LR (The integrated model: 0.88 ± 0.03 vs. EHR-only: 0.84 ± 0.04 ; KG-only: 0.75 ± 0.02) and RF (The integrated model: 0.86 ± 0.02 vs. EHR-only: 0.82 ± 0.03). Notably, models relying solely on environmental EHR factors exhibited a significant performance boost when enhanced with KG embeddings. XGBoost improved from 0.63 ± 0.03 (EHR-only) to 0.75 ± 0.02 , reflecting a gain of 0.12. Similarly, SVM’s EHR-only performance (0.57 ± 0.06) was relatively weak but improved significantly to 0.78 ± 0.03 after incorporating KG embeddings.
- **Bipolar Disorder:** XGBoost achieved an AUC of 0.88 ± 0.02 , outperforming EHR-only (0.85 ± 0.02) and KG-only (0.66 ± 0.03) approaches. LR demonstrated the highest relative improvement (+0.17), increasing from 0.70 ± 0.03 (KG-only) to 0.87 ± 0.02 (integrated model), highlighting the benefits of multimodal integration.
- **Anxiety Disorders:** The integrated model consistently improved AUC across all classifiers (range: 0.81–0.83), compared to KG-only (0.65–0.69) and EHR-only (0.78–0.79). XGBoost performed best (AUC: 0.81 ± 0.01 vs. 0.78 ± 0.01 for EHR-only). The non-medical EHR-only model for anxiety performed the weakest among the three conditions, suggesting that lifestyle, environmental, and familial factors may contribute differently to anxiety disorders compared to MDD and bipolar disorder.

Additionally, our findings indicate that integrating medical KG embeddings helps prevent performance degradation in feature-limited scenarios, such as when using only environmental factors for MDD.

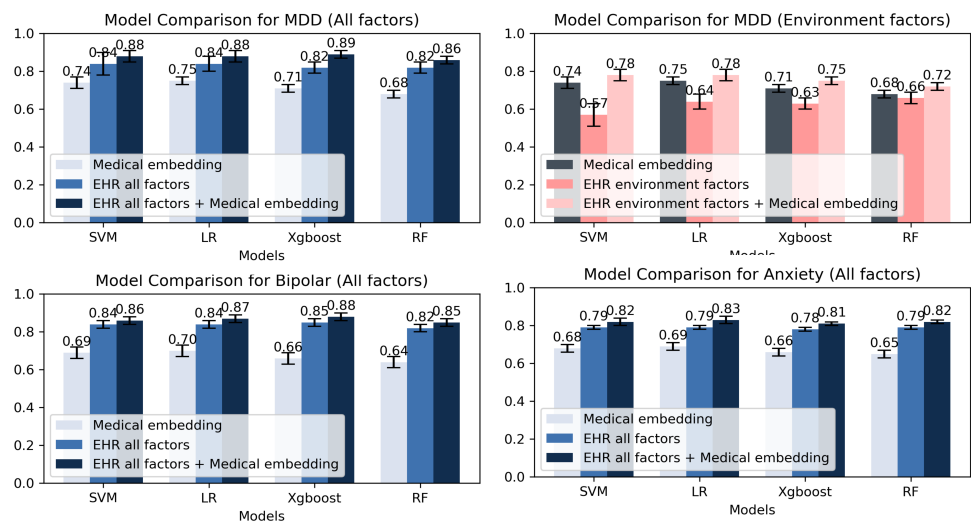


Fig. 2 Comparison of prediction performance across different diseases and models.

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Response for Reviewer 3

1. Why have the authors limited the extraction to just PubMed abstracts and not included the entire articles from the PubMed Central corpora? There seems to also be a discrepancy in the number of abstracts used, since the discussion mentions 500, but the online methods section mentions 234,000+ articles.

Response for the choice of PubMed abstracts rather than full-text articles:

We sincerely appreciate this insightful question regarding our data selection strategy. Our decision to focus on PubMed abstracts rather than full-text articles from PubMed Central was carefully considered based on three key factors:

- 1. Data Accessibility Constraints:** While PubMed Central (PMC) contains valuable full-text articles, approximately 70% of its content remains under copyright restrictions. By utilizing PubMed abstracts - which are uniformly available for over 37 million biomedical articles - we ensured complete reproducibility and equitable access to all data sources used in this study.
- 2. Relevance and Information Density:** Abstracts typically contain concise and structured summaries of key findings, making them highly effective for information extraction. While full-text articles may offer richer content, they also introduce challenges such as redundancy, variable structure, and increased noise, which can complicate analysis.
- 3. Computational Feasibility:** Processing full-text articles would require substantially more computational resources and would introduce additional complexity in natural language processing, such as handling extensive non-relevant sections (e.g., methods, discussions). Given the focus of our study, abstracts offered a balanced solution between coverage and computational efficiency.

We acknowledge that full-text extraction could provide additional insights, and we plan to explore this avenue in future research. We appreciate you raising this important point.

Response the discrepancy in the number of abstracts used: We apologize for any confusion caused by the discrepancy in the numbers mentioned. The 500 abstracts refers specifically to the manually annotated dataset we used for training and evaluation. In contrast, the 234,000+ abstracts mentioned in the online methods section represent the larger corpus of PubMed abstracts from which we extracted knowledge to construct our final knowledge graph. Thus, the 500 abstracts are a subset of the full 234,000+ abstracts used in the knowledge extraction process.

35 *2. Why have the authors focused on the use of PheCode over ICD-10? The authors*
36 *include a line in the manuscript, however it needs more explanation. For an application*
37 *of this research, would not prediction on ICD-10 codes be more meaningful?*

38 **Response:** We appreciate this insightful question regarding our rationale for using
39 PheCode over raw ICD-10 codes. While the UK Biobank indeed provides disease
40 records using ICD-10 codes, our adoption of PheCode mappings was motivated by two
41 key methodological and clinical considerations:

42 • **Reduction of Model Complexity & Enhanced Clinical Coherence:** ICD-10
43 contains 12,027 diagnosis codes, offering highly granular classifications (e.g., dis-
44 tinguishing between "Recurrent depressive disorder, current episode mild" [F33.0]
45 and "recurrent depressive episode in remission" [F33.4]). While this level of detail is
46 clinically valuable, it introduces significant sparsity and noise for machine learning
47 models.

48 The UK Biobank includes 6,245 recorded ICD-10 codes, of which 93.24% can
49 be mapped to corresponding PheCodes, yielding approximately 1,560 unique
50 PheCodes[1]. This hierarchical aggregation reduces feature dimensionality by 75%
51 while preserving critical clinical semantics, enabling models to identify robust
52 disease patterns rather than overfitting to coding variations.

53 • **Emphasis on Phenotypic Signatures over Etiological Specificity:** PheCode
54 focuses on clinically observable manifestations rather than etiological or admin-
55 istrative distinctions[2]. ICD-10 classifies depressive disorders into highly specific
56 subtypes, such as "single depressive episode, mild" (F32.0), "recurrent depressive
57 disorder, severe without psychotic symptoms" (F33.2), and "recurrent depressive
58 disorder, in remission" (F33.4). While these classifications serve administrative and
59 billing purposes, they create fragmentation that can obscure the core phenotypic
60 signal of major depression.

61 In contrast, PheCode 296.22 ("Major depressive disorder") consolidates all clini-
62 cally significant depressive episodes into a unified phenotype (see Table 1). This
63 phenotype-driven harmonization allows models to focus on biological and behavioral
64 hallmarks of depression—such as anhedonia and sleep disturbances—rather than
65 transient severity states or billing-driven distinctions. This approach is particularly
66 crucial in mental health research, where symptom overlap (e.g., between depression
67 and anxiety) and diagnostic heterogeneity pose major challenges.

68 We have incorporated this discussion into the main text, with the relevant additions
69 highlighted in red.

70 **[Change Made in Main Text 2.5 Enhancing MDD Prediction through**
71 **Knowledge Graph Embeddings]**

72 Additionally, we selected PheCodes to enhance clinical relevance and model robustness
73 by consolidating granular ICD-10 classifications into clinically meaningful pheno-
74 types, reducing dimensionality by approximately 75% while prioritizing core disease
75 manifestations over administrative distinctions. This approach mitigates sparsity and
76 diagnostic inconsistencies in mental health disorders, aligning with PheWAS objectives

Table 1 ICD-10 Codes Corresponding to PheCode 296.22

ICD-10 Code	Description
F20.4	Post-schizophrenic depression
F32.8	Other depressive episodes
F33.0	Recurrent depressive disorder, current episode mild
F33.1	Recurrent depressive disorder, current episode moderate
F33.2	Recurrent depressive disorder, current episode severe without psychotic symptoms
F33.3	Recurrent depressive disorder, current episode severe with psychotic symptoms
F33.4	Recurrent depressive disorder, currently in remission
F33.8	Other recurrent depressive disorders

to capture broad phenotypic patterns in EHR data. By emphasizing symptom-driven categorizations, PheCodes improve model generalizability and interpretability, overcoming the transient severity variations inherent in ICD-10 classifications [1, 2].

3. The authors have mentioned that the MDKG is more extensive than its counterpart knowledge graphs (e.g., PrimeKG or SPOKE), however, having more relations in a graph does not necessarily mean higher quality or usefulness. It would have been useful to: — Either see the added benefit of using MDKG for prediction tasks over BioBank, when compared with knowledge embeddings from SPOKE or PrimeKG or GENA (i.e., SVM + GENA AUROC). — OR do a human evaluation of random relations from MDKG vs GENA vs PrimeKG

Response: We appreciate the reviewer’s suggestion. In response, we have replaced the original comparative analysis of PrimeKG, MDKG, and GENA with a human evaluation, as requested. We have made corresponding modifications in the main text to reflect these changes.

[Change Made in Main Text 2.4 Human Evaluation of Triplet Quality Across Knowledge Graphs]

We conducted a comparative analysis of MDKG against two large-scale general biomedical knowledge graphs—PrimeKG [3] and SPOKE [4]—as well as a mental health-specific KG, GENA [5]. Our focus was on the density of directly connected information relevant to MDD, anxiety, bipolar disorder, schizophrenia, and depression. The analysis revealed that MDKG contains 5 to 50 times more information on these mental disorders than the comparative KGs, as shown in Figure ??d, highlighting its superior coverage and comprehensiveness.

To further assess MDKG’s quality, we conducted a human evaluation of randomly selected relationships from PrimeKG, MDKG, and GENA. Four evaluators—two psychiatrists (d1, d2) and two medical PhDs (phd1, phd2)—assessed the correctness and relevance of the extracted triplets. Given the varying information richness of the three knowledge graphs, we provided the maximum available data for each: triplets, PMIDs, and sentences for GENA; triplets and database sources for PrimeKG; and triplets,

107 PMIDs, sentences, journal, impact factor, frequency, and date for MDKG. For eval-
 108 uation, we selected two diseases, schizophrenia and anxiety disorders, and randomly
 109 sampled 100 non-redundant triplets per disease. Each evaluator reviewed the same set
 110 of examples, ensuring a consistent and unbiased comparison.

111 Evaluation metrics included correctness (whether the triplet was correct based on
 112 the provided sentence), insightfulness (rated 1–3 for novelty, reliability, and useful-
 113 ness), and evaluation time (average time for 200 examples). Correctness was excluded
 114 for PrimeKG, as it lacks original sources. Data analysis used Cohen’s Kappa for inter-
 115 rater consistency, with values ranging from -1 (no agreement) to 1 (perfect agreement),
 116 and the agreement rate between evaluators. Following is a summary of our findings:

- 117 • **Comparative Analysis of Triplet Distributions:** Overall, for triplets related to
 118 anxiety and schizophrenia, the relation and entity architecture across the knowledge
 119 graphs demonstrate distinct graph differences, as shown in [Figure 1](#). PrimeKG pre-
 120 dominantly contains generic associations, with approximately 90% of the relations
 121 labeled as "associated with," mostly involving gene entities. In contrast, MDKG
 122 exhibits more clinically actionable relationships, including approximately 20% "ther-
 123 apeutic relationships," 11% "risk relationships," and 10% "diagnostic relationships."
 124 GENA’s relationship distribution demonstrates linguistic specificity but clinical
 125 fragmentation, with approximately 85% of relations occurring 5 times. As a result,
 126 many of these relations lack standardization, such as "RECEIVE APPROVAL BY
 127 U.S. FOOD AND DRUG ADMINISTRATION FOR TREATMENT OF."
- 128 • **Quality Metric Performance:** For correctness ([Figure 2b](#)), MDKG achieved an
 129 accuracy of 79.0% ($\text{SEM} = \pm 0.07\%$), slightly outperforming GENA, which reached
 130 78.3% ($\text{SEM} = \pm 0.54\%$). The errors in MDKG were primarily attributed to semantic
 131 ambiguities (28%), rather than factual inaccuracies (12%). In terms of insightfulness,
 132 PrimeKG (2.09 ± 0.01) and MDKG (2.18 ± 0.02) scored lower than GENA (2.29
 133 ± 0.02). This difference likely reflects evaluators’ emphasis on the novelty of knowl-
 134 edge as a key metric. While PrimeKG and MDKG offer more extensive knowledge
 135 integration—especially MDKG, which incorporates clinical consensus data dating
 136 back to 1985—24.34% of MDKG’s schizophrenia-related triplets were derived from
 137 studies published before 2010. Evaluators frequently described these entries as "con-
 138 solidated textbook knowledge" in their post-evaluation feedback. This observation
 139 highlights the challenge of the need to balance evidence maturity (strength) with
 140 temporal relevance (novelty) in KG development.
- 141 • **Inter-Rater Reliability Assessment:** We observed significant intra-group vs
 142 inter-group discrepancies in evaluation consistency ([Figure 2e](#)). Clinical experts
 143 (d1-d2) achieved moderate agreement in correctness assessment (kappa: 0.43,
 144 Agreement: 76.7%), substantially higher than cross-disciplinary pairs like d2-phd1
 145 (kappa: 0.04). Similarly, researchers (phd1-phd2) showed strong consensus in insight-
 146 fulness evaluation (kappa: 0.63, Agreement: 80.0%), whereas clinician-researcher
 147 pairs had a kappa value of only 0.28. This pattern suggests divergent evaluation
 148 priorities—clinicians may have prioritized therapeutic validity (e.g., contraindica-
 149 tions, treatment effects), while researchers placed greater emphasis on mechanistic
 150 explanations.

- **Time Cost of Evaluation:** MDKG’s enriched contextual features significantly reduced evaluation time by 62–77% compared to PrimeKG and GENA (Figure 2c). Clinicians spent an average of 31 ± 2.24 seconds per MDKG triplet, compared to 129 ± 11.51 seconds for PrimeKG and 85.25 ± 4.66 seconds for GENA. This reduction in evaluation time was strongly correlated with metadata completeness, highlighting MDKG’s efficiency in facilitating rapid and informed assessments.
- **Validation of Confidence Metric:** Our confidence score exhibited a strong positive correlation with human insightfulness ratings (mean correlation: 0.78, std: 0.00), indicating that the factors used in its design closely align with the criteria humans consider when assessing the value of a triplet. This strong alignment further validates the robustness and reliability of our approach.

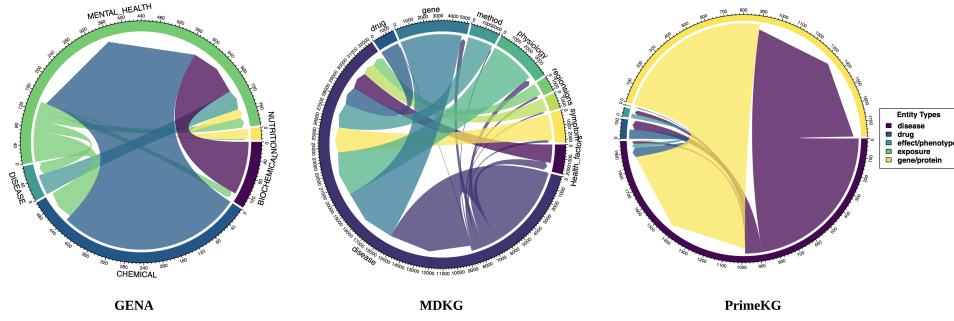


Fig. 1 Connections between different entity categories across different knowledge graphs.

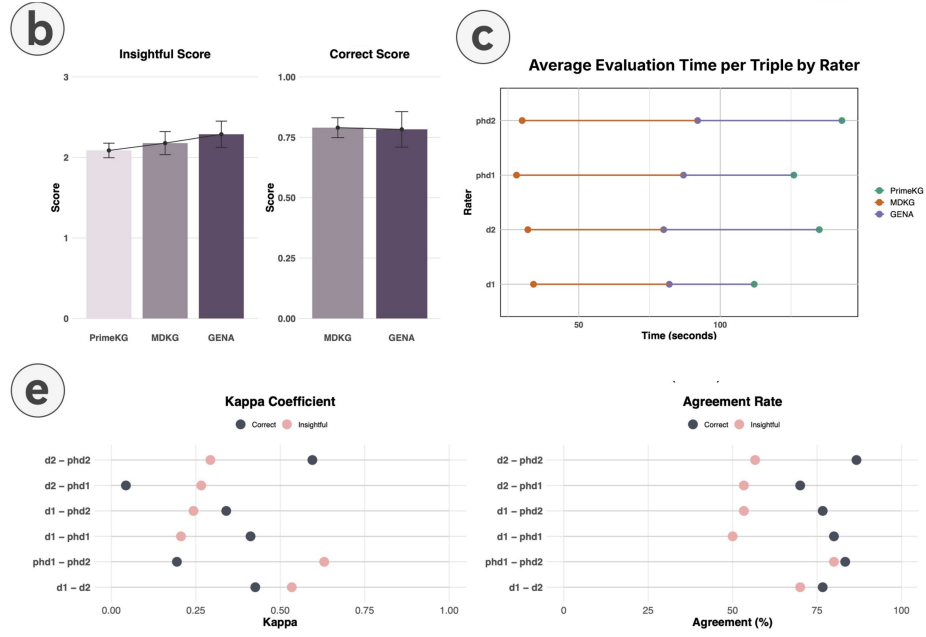


Fig. 2 The Manual check Results.

162 4. Why have the authors used QuickUMLS over cTAKES or MetaMap?

163 **Response:** Thank you for your insightful question regarding our choice of QuickUMLS
 164 over other tools such as cTAKES or MetaMap. In our study, both the entity annotation
 165 and entity linking stages require identifying the most similar UMLS term for a given
 166 noun phrase or entity. Ideally, this could be achieved by computing the embedding
 167 similarity between the given phrase or entity and all terms in UMLS. However, due
 168 to the vast number of terms in UMLS, this approach is computationally impractical.

169 To address this challenge, we adopt a two-step approach: (i) **Candidate**
 170 **Retrieval:** QuickUMLS is used to retrieve a candidate term set from UMLS that is
 171 likely to be similar to the given noun phrase or entity. (ii) **Similarity Computation:**
 172 We then compute the embedding similarity between the given noun phrase or entity
 173 and each term in the candidate term set, selecting the most similar term as the final
 174 linking target.

175 • **Efficient Fuzzy Matching with Improved Recall:** QuickUMLS enables fuzzy
 176 matching, effectively handling misspellings, morphological variations, and non-exact
 177 term matches with high accuracy. Given our objective of maximizing recall to ensure
 178 comprehensive concept mapping, QuickUMLS emerges as the most suitable choice
 179 among available tools, including cTAKES and MetaMap. As shown in Table 3,

QuickUMLS achieves the highest recall (0.75 at $\alpha = 0.6$), outperforming cTAKES (0.53) and MetaMap (0.48).

- **Computational Efficiency for Large-scale Datasets:** QuickUMLS is optimized for high-throughput processing, enabling efficient entity linking in large-scale datasets. Compared to cTAKES and MetaMap, it achieves significantly faster processing speeds (Table 3). At $\alpha = 1.0$, QuickUMLS processes documents in 143 ms per document, markedly faster than cTAKES (3,852 ms) and MetaMap (19,295 ms).

Given these advantages, QuickUMLS provides an optimal balance of efficiency, recall, and scalability, making it the most suitable choice for our study.

Method		Prec	Rec	F-1	ms/doc
MetaMap		0.49*	0.48*	0.48*	19,295*
cTAKES		0.71	0.53*	0.61	3,852*
QuickUMLS	$\alpha = 0.6$	0.50*	0.75	0.60	1,594*
	$\alpha = 0.7$	0.60*	0.66*	0.63	680*
	$\alpha = 0.8$	0.63*	0.60*	0.61	332*
	$\alpha = 0.9$	0.64*	0.56*	0.60	193*
	$\alpha = 1.0$	0.67*	0.54*	0.60	143

Table 1: Results for the i2b2 dataset. cTAKES outperforms QuickUMLS in precision, but QuickUMLS has better recall. QuickUMLS is 2.5 to 135 times faster than MetaMap or cTAKES. * indicates statistically significant differences from best value (Welch’s t -test, $p < 0.05$).

Table 1: Results for the i2b2 dataset. cTAKES outperforms QuickUMLS in precision, but QuickUMLS has better recall. QuickUMLS is 2.5 to 135 times faster than MetaMap or cTAKES. * indicates statistically significant differences from best value (Welch’s t -test, $p < 0.05$). [Collapse](#)

Fig. 3 Comparison of Recall and Processing Time Among QuickUMLS, cTAKES, and MetaMap Based on Data from [6]

5. Regarding SHAP analysis: Is there a way to indicate what is captured in certain embedding dimensions to be determined of higher predictive power? Section 2.5.2 Scenario 3 must indicate the AUROC over all variables + KG embeddings. The AUROC for SVM (all variables) + KG and LR (all variables) + KG are mentioned in the figure, but not in the results section. I am curious why the authors have focused on SVM and LR models instead of models such as XGBoost or Random forests?

Response for Missing AUROC for SVM and LR Models with KG Embeddings: Thank you for pointing out the missing AUROC values for the SVM and LR models with KG embeddings in the results section. We have now updated the manuscript to ensure consistency between the figures and the text. The AUROC for

199 the SVM + KG model is 0.88 ± 0.02 , and for the LR + KG model, it is $0.88 \pm$
200 0.02 . These values have been incorporated into the revised manuscript for clarity and
201 completeness.

202 **Response for Choice of SVM and LR Models over XGBoost and Random**
203 **Forest:** We selected SVM and LR models as our primary approaches due to their
204 interpretability and simplicity, which are particularly important in our study. Our
205 goal is to evaluate the contribution of non-medical EHR features to MDD prediction,
206 and both SVM and LR provide clear insights into feature importance, making them
207 well-suited for integrating KG embeddings.

208 However, we acknowledge that XGBoost and Random Forest are powerful mod-
209 els, particularly when dealing with complex, high-dimensional data and non-linear
210 relationships. While SVM and LR offer strong baseline performance and are easier
211 to interpret, we recognize that XGBoost and Random Forest can further enhance
212 predictive performance, particularly in more intricate datasets.

213 **Response for Inclusion of XGBoost and Random Forest Results:** Based on
214 your valuable feedback, we have now included results for both XGBoost and Ran-
215 dom Forest models in the revised manuscript. The AUROC for XGBoost with KG
216 embeddings for MDD prediction is 0.89 ± 0.02 , while for Random Forest, it is $0.86 \pm$
217 0.01 . These additions provide a more comprehensive evaluation of model performance
218 across different methodologies.

219 **Response for Inclusion of Anxiety and Bipolar Disorder Prediction Com-**
220 **parison:** In addition to expanding our analysis with new predictive models, we also
221 evaluated the performance of our method in predicting Anxiety and Bipolar Disorder.
222 We selected these conditions because Bipolar Disorder is often viewed as a combination
223 of Anxiety and MDD, and we aimed to assess how the same EHR variables and model-
224 ing approach perform across these mental health conditions. To ensure comparability,
225 we applied the same set of EHR factors used in our MDD prediction model.

226 The results indicate that the EHR factors-only model performs similarly for Bip-
227 olar Disorder as it does for MDD, with even slightly better performance when using
228 XGBoost (AUC: 0.82 vs. 0.85). However, for Anxiety, the EHR factors-only model
229 exhibited relatively lower performance (AUC 0.79), suggesting that the selected EHR
230 factors may not effectively capture key risk factors for Anxiety.

231 Overall, while different predictive models result in slight variations in AUC values
232 (differences 0.03 for each disease), the factors + medical embeddings model consis-
233 tently yields the best results. These findings further demonstrate the effectiveness and
234 utility of our approach in predicting multiple mental health conditions.

235 To reflect these updates, we have replaced Figure 4d with a revised version that
236 clearly displays the comparisons across different models, diseases, and EHR factors.
237 Additionally, based on these updated results, we have revised Section 2.5.2, "Assessing
238 the Impact of KG Embeddings on MDD Prediction," in the manuscript.

239 **[Change Made in Main Text 2.5.1 Integration of Knowledge Graph**
240 **Embeddings Enhances Predictive Accuracy for Mental Health Disorders]**

Our study presents a novel approach to representing a patient’s medical history using KG embeddings (see Figure 4g and Online Methods 4.4). Traditional models often encode diseases as dummy variables, which can lead to the curse of dimensionality, particularly in studies with limited sample sizes. For instance, representing 1,578 phenotypes typically requires an equally high-dimensional variable space, yet the available training samples may be insufficient to support such complexity. To address this, we introduce a KG embedding model that condenses a patient’s medical history into a 200-dimensional feature vector. This representation not only reduces the model’s dimensional burden but also integrates medical background knowledge from the knowledge graph, potentially improving diagnostic predictions and clinical insights.

Our analysis demonstrates that integrating medical KG embeddings with EHR-based factors significantly enhances prediction accuracy for MDD, bipolar disorder, and anxiety. Using ten-fold cross-validation across four classifiers (logistic regression [LR], random forest [RF], support vector machine [SVM], and XGBoost), we systematically evaluated three scenarios: (1) KG embeddings only, (2) non-medical EHR factors only, and (3) their integration. Performance was assessed using area under the ROC curve (AUC) with standard deviation, as shown in Figure 4.

- **Major Depressive Disorder:** When utilizing comprehensive non-medical EHR factors (see Supplementary for details), the integrated model (EHR + KG) achieved superior performance. XGBoost attained the highest AUC of 0.89 ± 0.02 , improving by 0.07 over the EHR-only model (0.82 ± 0.03) and by 0.18 over the KG-only model (0.71 ± 0.02). Similar trends emerged for LR (The integrated model: 0.88 ± 0.03 vs. EHR-only: 0.84 ± 0.04 ; KG-only: 0.75 ± 0.02) and RF (The integrated model: 0.86 ± 0.02 vs. EHR-only: 0.82 ± 0.03). Notably, models relying solely on environmental EHR factors exhibited a significant performance boost when enhanced with KG embeddings. XGBoost improved from 0.63 ± 0.03 (EHR-only) to 0.75 ± 0.02 , reflecting a gain of 0.12. Similarly, SVM’s EHR-only performance (0.57 ± 0.06) was relatively weak but improved significantly to 0.78 ± 0.03 after incorporating KG embeddings.
- **Bipolar Disorder:** XGBoost achieved an AUC of 0.88 ± 0.02 , outperforming EHR-only (0.85 ± 0.02) and KG-only (0.66 ± 0.03) approaches. LR demonstrated the highest relative improvement (+0.17), increasing from 0.70 ± 0.03 (KG-only) to 0.87 ± 0.02 (integrated model), highlighting the benefits of multimodal integration.
- **Anxiety Disorders:** The integrated model consistently improved AUC across all classifiers (range: 0.81–0.83), compared to KG-only (0.65–0.69) and EHR-only (0.78–0.79). XGBoost performed best (AUC: 0.81 ± 0.01 vs. 0.78 ± 0.01 for EHR-only). The non-medical EHR-only model for anxiety performed the weakest among the three conditions, suggesting that lifestyle, environmental, and familial factors may contribute differently to anxiety disorders compared to MDD and bipolar disorder.

Additionally, our findings indicate that integrating medical KG embeddings helps prevent performance degradation in feature-limited scenarios, such as when using only environmental factors for MDD.

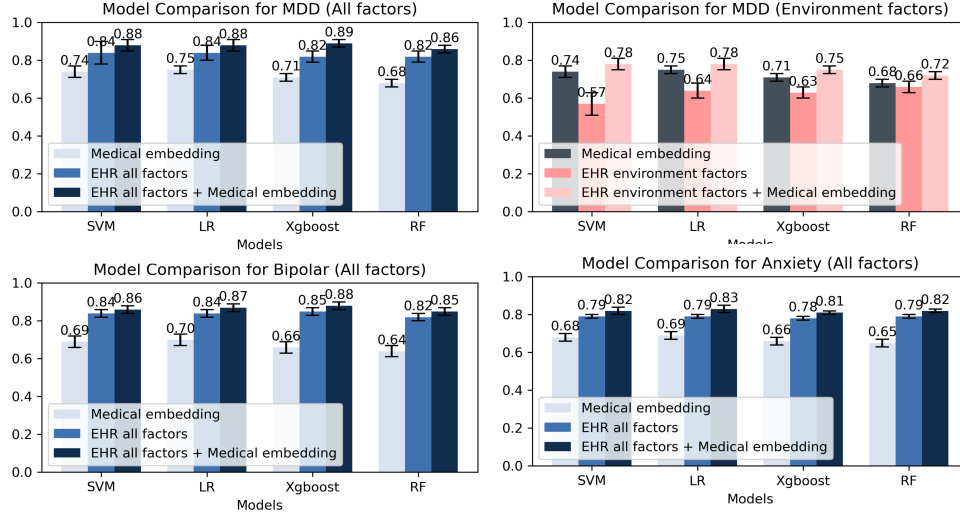


Fig. 4 Comparison of prediction performance across different diseases and models.

284 6. Regarding SHAP analysis: Is there a way to indicate what is captured in certain
 285 embedding dimensions to be determined of higher predictive power?

286 **Response:** We appreciate the reviewer’s insightful question. In our study, the
 287 embedding dimensions do not have fixed semantic meanings, as they are learned
 288 representations that depend on the training process, initialization, and data distribu-
 289 tion. Unlike explicitly defined features, embeddings capture latent relationships in an
 290 abstract manner, making their individual dimensions not directly interpretable.

291 Regarding SHAP analysis, while it provides valuable insights into which embed-
 292 dings contribute more significantly to the predictive task, it does not assign
 293 concrete meanings to specific dimensions. Given the inherent instability of embed-
 294 dings—meaning that feature correspondence across different training runs is not
 295 fixed—interpreting individual dimensions remains a challenge.

296 We acknowledge the importance of interpretability in embedding-based models and
 297 greatly appreciate the reviewer’s suggestion.

298 7. Have the authors used ChatGPT interface or the underlying API (e.g., GPT 3.5)?
 299 Were there any differences between ChatGPT and GPT-4o? What are the added ben-
 300 efits for entity annotation using QuickUMLS that is not captured through the use of
 301 LLMs?

302 We sincerely appreciate the reviewer’s insightful question.

303 **Response for the differences between ChatGPT and GPT-4o:** In this study,
 304 we utilized the APIs of both GPT-4 and ChatGPT, with the primary differences being

task accuracy and cost. For tasks requiring high precision and smaller data volumes, such as data annotation, we used GPT-4 due to its superior accuracy. Conversely, for tasks involving larger data volumes and simpler objectives, such as extracting population characteristics from tables or retrieving study details from abstracts, we opted for ChatGPT, which offers a more cost-effective solution without compromising efficiency.

Response for the Role of QuickUMLS: While BERN-2 pre-annotated most common biomedical categories (e.g., diseases, genes, drugs), additional processing was required for categories not covered by BERN-2 or those with lower annotation accuracy (e.g., procedures, lifestyle factors, anatomical structures).

To address this, we employed GPT-4 to extract noun phrases missed by BERN-2, which were then processed using QuickUMLS to assign entity types to the identified noun phrases. During the entity annotation process, QuickUMLS offers two key advantages:

- **Comprehensive Coverage of Semantic Types:** QuickUMLS includes a broader range of entity types beyond those recognized by BERN-2, a biomedical-specialized NER tool. Notably, it covers entities related to Health Factors, Method/Procedure, and Sign, which are often overlooked in biomedical NER processes.
- **Higher Precision:** In the pre-annotation process, our priority is to minimize incorrect annotations, as correcting them requires additional human effort and is often more time-consuming than directly assigning the correct entity type. Since UMLS provides relatively accurate semantic type assignments, directly linking entities to UMLS improves annotation precision. Compared to allowing GPT-4 to generate entity types, linking offers greater controllability in ensuring accuracy. By setting a high embedding similarity threshold, we can further ensure precise linking. Additionally, our experiments show that GPT-4’s NER performance is suboptimal for entity types such as Health Factors, Method/Procedure, and Sign—categories with unclear boundaries—achieving a precision of approximately 0.8.

This approach enhances annotation accuracy while reducing manual correction effort, ensuring a more reliable and efficient entity annotation process.

8. Capturing contextual information in knowledge graphs is not entirely novel has been a topic that has been covered in other publications. What is the underlying model used by the authors (e.g., RDF, reification)*

Response: We appreciate the reviewer’s insightful question. Our approach aligns more closely with RDF, as we structure our knowledge graph by enriching triplets with contextual attributes rather than employing full reification. However, our key innovation lies in the types of contextual features we extract (Table 2), which distinguish our method from traditional approaches. Specifically, we enhance knowledge representation by incorporating:

- **Conditional Statements:** Traditional knowledge graphs (KGs) typically extract plain factual triples (e.g., Depression associated with FMS), often omitting

population-specific or experimental conditions. We explicitly capture such conditions to improve precision and reliability (e.g., Depression associated with FMS [only in women]), ensuring that relationships are contextualized rather than overly generalized.

- **Study Population Characteristics:** We integrate demographic and cohort attributes (e.g., age, gender, racial composition, sample size) as triplet-level attributes. These details are extracted from the full text of articles, particularly from baseline characteristics tables and abstracts. This is crucial because conclusions drawn from one study may not generalize across different populations due to demographic differences.

- **Source and Confidence Metadata:** Instead of treating all extracted knowledge as equally reliable, we append source provenance (e.g., PubMed ID, first author, publication type, date) and confidence assessments (e.g., frequency, confidence score, and statistical methods used) to evaluate the credibility of extracted knowledge.

Compared to traditional RDF-based approaches, which primarily focus on structural representations, our method emphasizes biomedical-specific contextual information, ensuring that medical knowledge extraction and application are both accurate and clinically relevant.

Category	Contextual Features
Conditional Statements	occurs in, located in
Baseline Characteristics	Average age, Gender distribution, Racial composition, Educational attainment, Employment status, Overall sample size, and Cohort name
Source Information	Source, Authors, Year, Journal, DOI, Pmid, sentence, impact
Confidence	Confidence score, frequency, statistical analysis method, publication type

Table 2 Contextual Features for MDKG

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1 Response for Reviewer 1

2 *The authors have addressed my concerns adequately. I don't have further comments.*
3 *Tutorials and demos would be beneficial. A UI interface like gradio would be useful*
4 *and increase the impact. A python API would also be nice.*

5 **Response:** Thank you for your thoughtful feedback and encouraging remarks. We're
6 glad our revisions have addressed your concerns.

7 You suggested adding tutorials, demos, a Gradio-style UI, and a Python API to
8 enhance utility and impact. We're pleased to report that our web interface has already
9 been updated with two new modules:

- 10 1. **Entity-query suggestions** to guide users during searches.
- 11 2. **Path-discovery** for exploring multi-hop relationships between biomedical entities
12 (see Figure 1).

13 These enhancements improve usability and knowledge exploration, and we will con-
14 tinue refining them. Tutorials, interactive demos, and a Python API are on our
15 development roadmap and will be released soon to foster broader adoption and
16 integration.

17 Thank you again for your constructive suggestions.

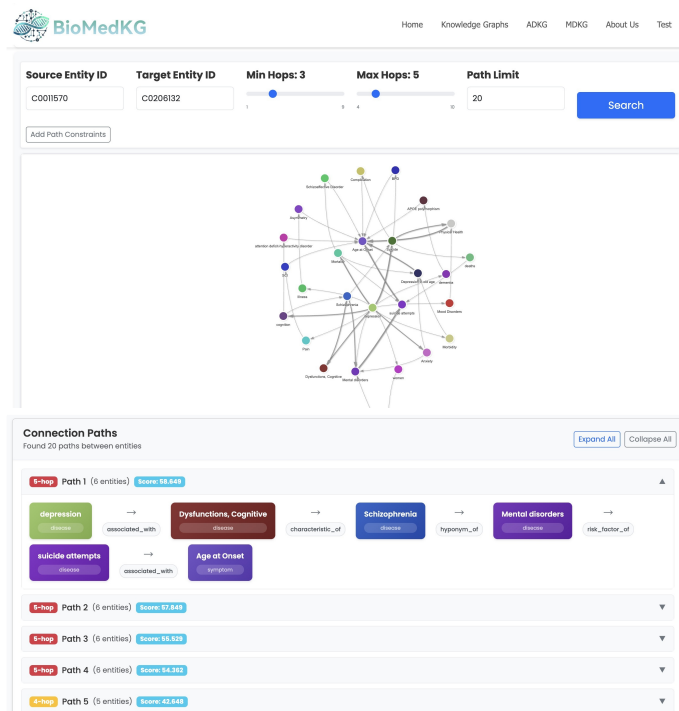


Fig. 1 The Path Discovery Module

Response for Reviewer 2

1. *Usage of Phecodes vs. ICD-10: The explanation provided for choosing phecodes over ICD-10 codes remains unconvincing. The authors argue phecodes avoid confusion with billing practices inherent to ICD-10. However, a simpler approach to reducing entities and complexity could be using the first two digits of ICD-10 codes, thereby achieving the intended simplification without introducing an entirely different coding system. The authors should provide a clearer justification for preferring phecodes specifically.*

Response: Thank you for this insightful suggestion. While truncating ICD-10 codes to their first two digits can reduce granularity, we chose Phecodes because they offer clinically curated groupings and proven performance in EHR research—benefits that simple truncation cannot provide.

a. Phecodes—developed and refined by clinical experts—aggregate ICD-9/10 codes into coherent, clinically relevant phenotypes. Unlike crude two-digit ICD-10 truncation, which can lump together unrelated or overly vague conditions, Phecodes group only genuinely related diseases, thereby improving model discrimination and interpretability [1–3]. Below, we illustrate this advantage for *major depressive disorder (MDD)*, *bipolar disorder*, and *anxiety disorders*:

1. Major Depressive Disorder (MDD)

Rather than a single ICD-10 code, MDD is distributed across multiple categories:

- F32.x: Single depressive episodes (varying severity and psychotic features)
- F33.x: Recurrent depressive disorders
- F20.4: Post-schizophrenic depression (clinically managed like MDD)

If only the first two digits are used, these relevant conditions are split across F32, F33, and F20, leading to fragmentation or omission. By contrast, Phecode 296.22 unifies all relevant ICD-10 codes into one cohesive phenotype, ensuring consistent cohort selection and richer feature representation.

2. Bipolar Disorder

ICD-10 includes bipolar disorder under F31, but related manic states are listed under F30, including:

- F30.x: Hypomania, mania with/without psychotic symptoms
- F32.8: Other depressive episodes frequently recorded in bipolar contexts

Simply truncating to F31 omits manic presentations, whereas Phecode 296.1 correctly merges F30, F31, and other related depressive codes to reflect the full spectrum of bipolar affective disorder.

3. Anxiety Disorders

The ICD-10 structure for anxiety is fragmented:

- F40.x: Phobic anxiety disorders (e.g., agoraphobia, social phobia)
- F41.x: Generalized anxiety, panic disorder
- F06.4, F93.1, F93.2: Organic and childhood anxiety subtypes

57 Using only F40 or F41 would omit important anxiety subtypes. Phecode 300.1 uni-
58 fies these diverse yet related ICD-10 codes into a single phenotype, supporting more
59 complete and clinically valid representations of anxiety for downstream analysis.

60 b. Phecodes have been rigorously validated and are widely used across major EHR
61 research consortia—including eMERGE, All of Us, and UK Biobank [4]. Their utility
62 spans numerous applications, from PheWAS to phenotype risk scoring, demonstrating
63 robust performance in diverse clinical contexts [5, 6].

64 Thank you for this insightful comment. We trust that our detailed rationale clarifies
65 why Phecodes are preferable to truncated ICD-10 codes for defining mental health
66 phenotypes.

67 *2. Choice of ChatGPT (Closed-source) Over Open-source Alternatives: The authors*
68 *still have not sufficiently clarified their rationale for selecting ChatGPT, a proprietary*
69 *closed-source model, over available open-source alternatives such as BERT or Llama.*
70 *Given the sensitivity of medical data and the importance of transparency, employing*
71 *a locally hosted or open-source system seems preferable. The authors should clearly*
72 *justify the benefit or necessity of their current choice.*

73 **Response:** Thank you for raising this important point. To clarify, our use of Chat-
74 GPT is confined to processing publicly available biomedical literature (e.g., PubMed
75 abstracts) and does not involve any patient-level or sensitive data, so privacy concerns
76 are not applicable here.

77 During the initial triple-extraction phase from unannotated abstracts, we relied
78 on a locally hosted, fine-tuned BERT model. We then incorporated ChatGPT in
79 later stages because it delivered superior performance on complex relation-extraction
80 tasks and dramatically reduced development time. Although we evaluated open-
81 source alternatives (such as LLaMA), we found that achieving comparable accuracy
82 required extensive fine-tuning and significant computational resources—especially in
83 low-resource or high-complexity settings.

84 We share your commitment to transparency and reproducibility in medical AI.
85 Accordingly, we are actively exploring locally deployable open-source models (e.g.,
86 LLaMA, DeepSeek) for downstream tasks to maintain full control over model behavior
87 and ensure privacy.

88 We appreciate your suggestion and will continue to prioritize open-source, privacy-
89 preserving solutions in future work.

90 *3. Clarification of "Disease Prediction": The use of the term "disease prediction"*
91 *remains ambiguous. It is unclear if the authors are predicting future diagnoses or*
92 *simply identifying patients already labeled with Major Depressive Disorder (MDD) in*
93 *their dataset. The authors must explicitly clarify if their approach involves genuine*
94 *predictive modeling—predicting future onset of disease—or merely labeling existing*
95 *diagnoses accurately.*

96 **Response:** Thank you for this helpful comment. We'd like to clarify that our mod-
97 eling task truly predicts future disease onset, not merely identifies existing cases. To
98 do this, we defined cohorts for MDD, anxiety, and bipolar disorder and extracted
99 each patient's EHR data from 1–7 years before their first recorded diagnosis. All

features—both EHR-derived variables and knowledge-graph embeddings—are drawn solely from this pre-diagnosis window, ensuring no information from after diagnosis influences the model. This design mimics a real-world clinical screening scenario, where we flag individuals at elevated risk before they receive a formal diagnosis.

We appreciate the reviewer’s suggestion and have revised the manuscript to make this distinction clearer.

[Change Made in Main Text]

We constructed a case-control cohort focusing on three target conditions: major depressive disorder (MDD), anxiety, and bipolar disorder. For each case, we compiled the patient’s pre-diagnosis health profiles based on medical records from one to seven years prior to their first recorded diagnosis, aiming to capture early predictive signals.

4. Explanation and Usage of "Triplets": While "triplets" are now mentioned in the abstract, the manuscript lacks a specific and clear definition of what constitutes a "triplet" and the rationale behind their use. Providing a concise explanation in the methodology would enhance clarity and understanding of their importance within the knowledge graph.

Response: Thank you for this suggestion. We’ve now defined “triplets” more explicitly as ordered semantic units (head entity, relation, tail entity) that capture factual or inferred links between two biomedical concepts. For example, (Fluoxetine, treats, Major Depressive Disorder) denotes a drug-disease treatment relationship, and (TNF-alpha, associated with, inflammation) reflects a biomolecular association. These triplets are derived either from biomedical literature—using large language models for relation extraction—or from curated biomedical databases, and together form the edge set of our constructed knowledge graph.

We have now added a clear definition and explanation of triplets in the manuscript accordingly. Thank you again for highlighting this important point.

[Change Made in Introduction]

For instance, they can transform a statement like “The orbitofrontal-hippocampal pathway is crucial in mediating depression”^[7] into a structured format known as a triplet—a semantic unit in the form of (head entity, relation, tail entity)—such as [orbitofrontal-hippocampal pathway, associated with, depression], optionally enriched with additional contextual properties for precision.

5. Clarification Regarding AUC Values: The authors mention an improvement in AUC values (abstract) "from 0.02 to 0.21." This statement remains unclear and potentially misleading. Typically, an AUC (Area Under the Curve) value below 0.5 indicates a model performing worse than random chance, with meaningful AUC values typically ranging from 0.5 (random chance) to 1.0 (perfect classification). Thus, an AUC improvement to 0.21 is confusing and suggests either a typographical error or a conceptual misunderstanding. The authors should review and clarify this to avoid misinterpretation.

140 **Response:** Thank you for highlighting this ambiguity. We apologize for any con-
141 fusion—“from 0.02 to 0.21” describes the increase in AUC achieved by adding
142 knowledge-graph embeddings, not the absolute AUC values themselves. In other
143 words, across various models and disease categories, we observed AUC improvements
144 ranging between 0.02 and 0.21 over each model’s baseline performance. We have
145 updated the abstract and relevant manuscript sections to make this distinction clear.

146 **[Change Made in Abstract]**

147 Applied to predictive modeling in the UK Biobank, MDKG-enhanced representations
148 yielded significant gains in predictive performance (AUC improvement: 0.02–0.21)
149 across multiple mental disorders.

Response for Reviewer 3

While the SVM / LR results are acceptable, and the authors have also included results from XGBoost and Random Forest models in the recent iteration, I am confused on the claim around interpretability - primarily around the response for model selection and the SHAP analysis. That is, the following statement in the rebuttal: "We selected SVM and LR models as our primary approaches due to their interpretability and simplicity" is contradictory to another statement: "Given the inherent instability of embeddings—meaning that feature correspondence across different training runs is not fixed—interpreting individual dimensions remains a challenge." Why have the authors not pursued the use of deep learning models for this task when interpretability and justifications on the inference, are going to be impossible even with SVM / LR models?

Response: Thank you for this perceptive comment. You're correct that using inherently opaque embeddings with interpretable classifiers (SVM, LR) does not fully reveal which specific graph features drive predictions. To clarify:

1. Isolating embedding impact.

Our primary goal was to quantify the *performance gain* from adding knowledge-graph embeddings to disease prediction. By pairing the embeddings with simple, well-understood models (SVM and LR), we ensured that any accuracy improvement could be attributed to the embeddings themselves, rather than to a powerful but complex classifier.

2. SHAP as a group-level gauge.

We did not use SHAP to decode individual embedding dimensions—indeed, their semantics can shift across training runs. Instead, SHAP highlights the *collective* importance of the embedding block relative to conventional EHR features. In our results, embedding-derived features frequently rise to the top of the importance rankings, demonstrating their added value even if we cannot yet map each dimension to a specific biological concept.

We agree that fully interpreting these embeddings—or justifying predictions via a deep model—remains an open challenge. Exploring alternative architectures and embedding-explanation techniques is an important direction for future work. Thank you again for helping us clarify the scope and limitations of our interpretability claims. Furthermore, we have revised and enhanced our GitHub repository to improve clarity and usability, as per your helpful suggestion.

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